Signaling Quality or Gaming the System? Evidence from College-Major Accreditation

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This paper examines program-level accreditation in higher education, analyzing student responses and institutional adjustments to this quality signal. Using administrative data from Chile (2007-2018) and a difference-in-differences approach exploiting staggered adoption, we estimate the effects of this information shock. First-time accreditation increases student demand (applications by 10.2%, enrollment by 7.3%), attracts stronger students, and reduces transfers, suggesting improved match quality. These effects are consistent across socioeconomic backgrounds. Investigating the signal's construction, we find no evidence of pre-accreditation increases in long-term structural inputs but document a significant rise in on-time graduation rates concentrated immediately before evaluation and not sustained post-accreditation. Effects are amplified by institutional reputation and weaker under mandatory regimes. Our findings reveal a policy trade-off: accreditation signals can enhance student sorting, even when institutional responses focus on evaluation metrics rather than long-term, structural improvements.

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1 Introduction

Students choose college majors under considerable uncertainty about future earnings, costs, and educational quality (Dillon and Smith, 2020; Stange, 2012). In many systems, including Chile's, this decision is a choice between institutions, conditional on a desired field of study. Third-party certification can lead students toward better choices and help high-quality programs distinguish themselves from a crowded field (Akerlof, 1970; Dranove and Jin, 2010; Elfenbein et al., 2015).

We study accreditation as a quality signal in higher education markets (Hoxby, 2002; MacLeod and Urquiola, 2015). The value of accreditation, however, is not uniform. While elite institutions rely on established, long-term reputations, accreditation is most valuable for institutions competing below the top tier (Hörner, 2002; MacLeod et al., 2017; MacLeod and Urquiola, 2015). For these programs, a positive signal distinguishes them from the "left tail" of the quality distribution.

Accreditation can also introduce perverse incentives. Economic theory highlights that when certification relies on specific metrics, firms tend to emphasize inexpensive, easily manipulated dimensions that increase their likelihood of certification, while neglecting more important but harder-to-improve aspects. (Dranove and Jin, 2010; Jin, 2005; Spence, 1973). This has been measured in other industries, including restaurant hygiene grading or financial reporting, where firms must decide between substantive compliance and managing appearances (Jin and Leslie, 2003; Jin et al., 2021; Leuz et al., 2008). This possibility raises the question of what exactly accreditation rewards, highlighting that it may not only distort students' choices but also programs' choices on what to improve before going through the accreditation process.

We investigate three questions. First, how do students respond to accreditation? Because students choose programs under uncertainty, we hypothesize that accreditation reduces ex-ante uncertainty, affecting their choices (Dranove and Jin, 2010). We test two predictions: (1) accreditation increases student demand (applications and enrollment); and (2) by enabling better-informed initial choices, it improves the student-program match, measured by reduced program switching (Dillon and Smith, 2020). We then ask whether this response is similar across socioeconomic groups. This is motivated by evidence that information frictions can be particularly severe for students from disadvantaged backgrounds, and a simple, public signal like accreditation could potentially enhance accessibility compared to complex information environments (Dynarski et al., 2021; Hoxby and Turner, 2015).

Second, how does the context in which a program receives accreditation affect the informativeness of the signal? We explore this through two distinct margins. (1) We test competing hypotheses on the interaction between accreditation and pre-existing institutional reputation. If the signal acts as a substitute for reputation (Elfenbein et al., 2012, 2015), its impact will be greatest for programs at less reputable institutions. If it acts as a complement by providing information on different quality dimensions (Börgers et al., 2013; Colombo et al., 2019), its effect will be largest at more reputable ones. (2) We test the hypothesis that the effect of accreditation on student outcomes will be significantly weaker under a mandatory regime. This is because the mandatory framework eliminates the informative self-selection mechanism present in a voluntary regime (Jin and Leslie, 2003; Spence, 1973), potentially reducing the signal's informational content (Jin, 2005).

Third, what is the nature of the signal that students are responding to? We investigate how university programs respond to the incentives to get accredited. Do they make long-term, structural investments in inputs like faculty or physical infrastructure, or do they focus on improving more malleable metrics that align closely with the evaluation cycle? We hypothesize that programs will focus on improving more malleable and visible metrics that are explicit inputs in the accreditation assessment [1], such as on-time graduation rates, rather than undertaking long-term structural investments.

To answer these questions, we exploit the institutional features of Chile's higher education system from 2007 to 2018, which provide an ideal quasi-experimental setup for our study. The key to our identification strategy is that while a program's decision to seek accreditation is endogenous, the public announcement of the final decision acts as a largely unanticipated information shock for students. This uncertainty operates on multiple levels. First, programs voluntarily decide when to undergo the lengthy and variable evaluation process, leading to a staggered rollout of accreditation decisions across the market. Second, and crucially, the final outcome is ex-ante uncertain. Programs often expect longer accreditation terms than they ultimately receive, making the final decision by the agency a potential surprise (Barroilhet, 2019; Barroilhet et al., 2022). For prospective students making application decisions, and for currently enrolled students considering switching programs, the release

[1] Appendix table A.2 describes the criteria measured by the CNA and how we approach them using the available data.

of a new accreditation status, including its duration, is new information that is plausibly exogenous to other short-term factors influencing their choices. This staggered timing allows us to use a difference-in-differences framework to compare changes in outcomes for programs before and after their first accreditation to a control group of not-yet-accredited and never-accredited programs, thereby isolating the causal effect of the information signal. The setting is further strengthened by a centralized admissions platform that allows us to observe the ranked preferences over programs of approximately $950,000$ applicants, and by the existence of both voluntary and mandatory accreditation regimes, which allows us to distinguish between the effects of a signal that is earned from one that is merely required.

We show that students respond significantly to the accreditation signal. First-choice applications to newly accredited programs rise by 10.2%, enrollment increases by 7.3%, and the applicant pool becomes academically stronger. Furthermore, the effect is consistent across socioeconomic backgrounds, appearing just as strong for students from low-income backgrounds. This finding makes a key contribution to the economics of higher education by complementing the rich literature on information interventions (Bettinger et al., 2012; Hoxby and Turner, 2015; Jensen, 2010; Wiswall and Zafar, 2015). While those experiments show researcher-provided information can improve outcomes, our results demonstrate that a real-world, institutionalized signal can be an effective and scalable tool for reducing information frictions. By analyzing how students update their choices, we provide new evidence on the real-world impact of the dynamic learning and information frictions documented by Bordon and Fu (2015); Stinebrickner and Stinebrickner (2012) and most recently Arcidiacono et al. (2025).

The heterogeneity of these effects reveals further variation. Our analysis by institutional reputation uncovers a key finding about market selection: programs at different quality tiers exhibit different pre-treatment trends in student outcomes, suggesting they may seek accreditation for different strategic reasons (e.g., to consolidate market share or as a remedial tool, as in Cameron et al. (2023)) . For the outcomes where the parallel-trends assumption holds, we find evidence consistent with complementarity. For instance, the effect of the accreditation on reducing permanent exits is concentrated in enhanced and top-tier institutions, engaging with the literature on the interaction between formal and informal signals (Bar-Isaac and Tadelis, 2008; Elfenbein et al., 2012; MacLeod et al., 2017; MacLeod and Urquiola, 2015). Finally, we find the signal's effect is substantially weaker under mandatory accreditation regimes, underscoring that its value is shaped by the surrounding regulatory environment.

We find no evidence of major pre-accreditation investments in long-term structural inputs. Instead, we document a significant increase in malleable metrics, like on-time graduation rates, that is concentrated immediately before a program's first evaluation and is not sustained in the post-accreditation period. This temporal pattern provides novel empirical evidence of institutions adjusting specific evaluated metrics in close alignment with the certification cycle. This represents our primary contribution to the literature on quality disclosure, extending insights from foundational theory (Milgrom, 1981; Spence, 1973) and empirical work in other sectors (Dranove and Jin, 2010; Jin, 2005; Jin and Leslie, 2003; Vatter, 2023) to the complex domain of higher education (Cameron et al., 2023). While canonical studies like Jin and Leslie (2003) take the signal's content as given, we examine its creation, showing that the information students receive is an equilibrium outcome of these institutional choices.

Unlike most prior work on accreditation, which either treats it as an exogenous control variable or focuses on its impact on institutional finances or behavior (Burnett, 2022; Dinarte-Diaz et al., 2023; Hastings et al., 2016), our analysis centers on student demand responses (Cameron et al., 2023). The unique institutional setting in Chile, combined with rich administrative data, allows us to overcome many of the identification challenges that have limited previous work.

These findings point to a crucial policy trade-off. On one hand, the signal appears to be welfare-enhancing from a matching perspective. By guiding students of all socioeconomic backgrounds to programs where they are more likely to persist, accreditation improves allocative efficiency and broadens market participation. On the other hand, our evidence on institutions' short-term metric adjustments raises questions about the policy's effect on productive efficiency. If institutions substitute this metric management for long-term, structural investments, the signal may improve sorting without necessarily increasing the human capital produced. This highlights a key challenge for policymakers: designing quality assurance systems that provide clear signals without quietly incentivizing compliance focused on the evaluation metrics.

The remainder of this paper is organized as follows. After describing the institutional background in Section 2, the data in Section 3, and the empirical strategy in Section 4, we present our results. We begin in Section 5 with our main causal findings on how students respond to the accreditation signal. In Section 6, we explore the heterogeneity of these responses. Finally, Section 7 turns to the signal itself, presenting our descriptive analysis of the institutional behaviors that precede accreditation. Section 8 concludes.

2 Institutional Background

2.1 Higher Education in Chile

The Chilean university system includes 61 public and private institutions that offer three- to six-year bachelor's degrees across a wide range of academic fields. As of 2020, 45 of these universities participate in a centralized admission system, which includes institutions of public, private, and private-parochial origin. This centralized mechanism is generally associated with higher academic selectivity and prestige, while the remaining universities, primarily private, conduct their own admissions and tend to enroll students with lower average entrance scores.

The centralized system assigns students to programs using a student-proposing Deferred Acceptance Algorithm (DAA), based on a composite admission score. This score combines results from the national standardized test—the *Prueba de Selección Universitaria* (PSU)—with high school GPA and class ranking. Roughly 95% of high school graduates take the PSU annually. Applicants submit a ranked list of up to ten program choices, and assignments are made subject to capacity constraints at each program.

The nature of this application process is a key feature of the market. Unlike systems where students apply to a university and select a major later, applicants in Chile apply directly to a specific program (a college-major combination). This structure frames the student's decision as a choice between different institutions, conditional on having a desired field of study (e.g., choosing between 'Business at University A' and 'Business at University B'). This institutional setup makes our analysis of program-level accreditation particularly relevant, as it is a specific signal students can use to compare programs across universities. Furthermore, it highlights the central role of pre-existing university reputation as a competing, general signal, which motivates our test for whether these two signals act as substitutes or complements.

The academic year in Chile runs from March to December, aligning with the calendar year. Students take the national university entrance exam (PSU) at the end of each year, with results typically released between Christmas and New Year's Eve. The centralized application window opens in early January and lasts three to five days, after which students are matched to programs based on their ranked preferences and admission scores. In the final weeks of the year, prior to the release of PSU results, universities promote their programs through billboard campaigns, transit advertisements, online materials, and university fairs organized independently or jointly with other institutions.

This structure has two key implications for our empirical design. First, since enrollment decisions and academic exposure both occur within the same calendar year, the timing of treatment relative to the year of enrollment can be interpreted without ambiguity. A student enrolling in year t experiences their full first academic year in t. Second, because accreditation decisions are typically issued mid-year, they become visible only for cohorts entering in year $t+1$, and do not influence application or enrollment behavior in the year of issuance. This timing feature allows us to cleanly separate anticipatory behavior from post-treatment responses in the event-study design. Accreditation status, once granted, becomes part of the information environment that students face during the December–January application cycle, acting as a publicly visible, formal indicator of program quality.

While the application analysis in this paper focuses on universities within the centralized system, where students' program rankings are observable, the analysis of enrollment and persistence and graduation includes all universities, regardless of their admission mechanism. This broader inclusion allows us to evaluate the impact of accreditation across the full spectrum of Chilean higher education institutions.

2.2 Accreditation System

Chile's accreditation system was formalized through the 2006 Quality Assurance Act, establishing the National Commission of Accreditation (CNA) as the central authority. While the CNA oversees both institutional and program-level processes, this study focuses on program-level accreditation, which provides a major-specific signal crucial for student choice. Appendix A provides a detailed description of the accreditation process, evaluation criteria, and variable mapping.

Accredited programs receive a public quality signal, a status lasting from zero (denial) to seven years, based on compliance with standards across multiple dimensions. Accreditation is granted jointly for a program across all campuses where it is offered under a unified curriculum, ensuring a single treatment date for our empirical setup. Although program and institutional accreditation are correlated, there is substantial within-institution variation in program status.

While voluntary in general, accreditation is mandatory for specific fields (education and medicine), which we exclude from our main analysis. Outside mandated fields, programs typically pursue accreditation voluntarily as a strategic decision (Barroilhet, 2019), and endogenously decide if and when to apply. This leads to a staggered adoption of accreditation across the market, illustrated in Figure 1. This staggered timing provides the quasi-experimental variation for our difference-in-differences design.

Our empirical analysis focuses on first-time program accreditation events. The outcome is publicly disclosed, becoming an observable quality signal for prospective students during the admissions cycle.

3 Data and sample

To analyze how students respond to program accreditation, we construct a panel dataset at the program-campus-year level spanning from 2007 to 2018. The dataset links student-level administrative records with institutional and program-level data, covering the full universe of accredited universities in Chile.

3.1 Data on Programs and Institutions

Records from the National Commission of Accreditation (CNA) provide the universe of accreditation decisions for undergraduate programs and institutions. These data include the timing and outcome of each accreditation application, the number of years granted when successful, and the exact month and year in which the decision was communicated to the program.

To capture inputs into program quality and institutional investment, we use data from the Higher Education Information System (SIES), which compiles annual information submitted by universities to the Ministry of Education. These data include tuition fees, declared enrollment capacity, faculty headcounts and contract types, and physical infrastructure. Institutions are required to submit this information each year prior to the admissions cycle, and the data about tuition and capacity become publicly available before students submit their applications.[2]

Many of resource variables, such number of buildings and meters of infrastructure, are reported at the campus level. To allocate these inputs to the program level, we implement an enrollment-weighted allocation procedure that accounts for both program size and expected demand. Specifically, we multiply each campus-level resource by the share of first-year enrollment in a given program relative to total first-year enrollment at the campus. This program-level allocation is then normalized by the median first-year enrollment of the program over the study period, which helps avoid distortions caused by year-specific enrollment shocks and better captures expected cohort size. The resulting measure reflects the intensity of available resources per expected student. Outcomes reported directly at the program level and do not require further scaling. These outcomes are used log-transformed form as appropriate in the empirical analysis.

3.2 Data on Student Applications and Enrollment

We use two primary administrative sources to track student applications and academic progression. First, data from the Department of Evaluation, Measurement, and Educational Registry (DEMRE) provides complete records of students participating in Chile's centralized admissions test. These include national entrance exam scores (PSU), socioeconomic characteristics, and the rank-order lists submitted by applicants. These data allow us to observe the set of programs each student applies to, along with academic and demographic attributes that characterize the applicants.

Second, student-level enrollment records from the Ministry of Education track the full universe of first-time entrants to university programs. These records capture each student's program, campus, and year of initial enrollment, as well as subsequent academic outcomes, such as whether the student remains enrolled, transfers, or graduates.

[2] Information about tuition and capacity is submitted to the Ministry approximately three months before the national entrance exam, ensuring that students can observe program characteristics prior to choosing where to apply.

3.3 Panel Construction and Outcomes

Programs in the application and enrollment datasets are identified at the campus-specific level. That is, if university u offers program p across multiple regions, we observe a separate record for each campus-year combination. This structure allows us to construct geographically disaggregated panels and to follow cohorts of students within each program-campus unit over time.

Accreditation decisions, however, are made at the national level for each program, regardless of location. A single application includes all campuses where the program is offered, and the resulting accreditation decision applies uniformly.[3] To ensure consistency in treatment assignment, we replicate the program-level accreditation decision across all campus-year units where that program is observed.

Our baseline panel is therefore structured at the program-campus-year level. For each observation, we observe the program's accreditation status in that year, together with several outcomes. These include the number of first-choice applications received (log-transformed), the number of first-year enrollees (log-transformed), and cohort-level persistence outcomes. Persistence is measured as the share of a first-year cohort that either drops out, transfers to another institution, or exits the system entirely within a defined window. These outcomes are constructed using student-level data but are collapsed to the program-campus-year level and anchored to the cohort's year of entry. This timing reflects the information available to students at the point they make application and enrollment decisions.

Graduation outcomes are constructed differently from other student-level outcomes. Rather than anchoring outcomes to the year of enrollment, we align graduation measures to the expected year of degree completion. Specifically, we identify each first-year cohort and compute the share of students who graduate from the same program within the nominal duration plus one year, a definition of "on-time" graduation used by the CNA in its accreditation guidelines.[4] For example, in a four-year program, a student who enrolls in 2010 is considered on-time if they complete their degree by 2014.

We then assign this outcome to year $t + d$, where t is the student's year of entry and d is the official program duration.[5] This structure allows us to test whether students are more likely to graduate on

[3] All campuses included in the accreditation request undergo peer review and submit self-evaluation reports. The agency issues a unified accreditation decision that covers the full program. See Section 2 for details.

[4] This is the measure of "on-time" graduation stated by the CNA in their regulations to evaluate the results of a program's performance.

[5] In our setup, the academic year overlaps with the calendar year, so a student enrolled in year t completes their first full year within that same calendar year. This implies that the expected graduation year for on-time

time when accreditation status becomes observable near the point of expected degree completion. It also captures how programs may push graduation just prior to evaluation, as on-time graduation is an explicit input in accreditation assessments.

This timing structure allows us to test whether accreditation influences program behavior or student outcomes close to the point of degree completion. Since accreditation is publicly disclosed during the students' final years, both students and programs may respond around that time, either by adjusting graduation incentives or by accelerating the completion of pending degrees. While this approach does not follow a fixed cohort throughout its full trajectory, it captures variation in graduation timing at the point where institutional incentives and student outcomes allign.

3.4 Sample Restrictions

We define the estimation sample to reflect settings in which accreditation decisions are strategic and student responses are most likely to reflect informational updating. To that end, we restrict attention to "traditional" undergraduate programs, understood as those offered in an in-person, full-time format. These programs represent the standard choice for recent high school graduates. We exclude evening, distance-learning, and modular formats, which typically attract non-traditional student populations and operate in distinct market segments.

Our main analysis focuses on programs in fields where accreditation is voluntary. Programs in education, medicine, and odontology are excluded from the baseline estimation sample, as accreditation is legally mandated in these fields and may be perceived as regulatory compliance rather than a discretionary signal of quality. We reintroduce these programs in Section 6 to compare effects across voluntary and mandatory regimes.

Starting from the full universe of undergraduate programs observed between 2007 and 2018, we apply the following restrictions. We exclude non-traditional modalities, drop programs with fewer than seven years of continuous data, and remove those with missing values or extreme volatility in enrollment.[6] We also exclude programs offered by universities that did not maintain institutional accreditation throughout the period. These restrictions ensure that we retain programs with stable

completion is $t + d$, not $t + d + 1$.

[6]Extreme volatility is defined as year-over-year changes exceeding 2.5 standard deviations of the program's historical series.

operations, reliable student cohorts, and consistent exposure to accreditation as a quality signal.[7]

After applying these restrictions, we retain 1,526 traditional-format programs. Of these, 128 were excluded from our estimations because they received accreditation before the first year considered in our study. Finally, of the remaining 1,398 programs 1,046 belong to fields where accreditation is discretionary and form our primary estimation sample. Within this group, 729 programs received accreditation for the first time during the study period and comprise the treatment group; the remaining 317 programs never received accreditation and work as never-treated programs.

3.5 Treatment Definition and Summary Statistics

We define treatment as the first year in which a program receives accreditation. Our main treatment variable is a binary indicator that switches to one in the year of first accreditation and remains zero otherwise. Programs are only treated once during the study window, and reaccreditations are not considered in this version of the analysis.

The timing of treatment is based on the year in which the CNA's decision is issued. Since accreditation is typically granted mid-year—after applications and enrollment decisions have already been made—we do not expect direct effects on those outcomes in year $t = 0$. However, the CNA requires programs under evaluation to disclose their ongoing accreditation process in advance, including through public channels. This may enable some prospective students to update their beliefs about program quality even before the official decision is announced.[8] In contrast, already-enrolled students may respond immediately to news of accreditation through reduced dropout or transfer behavior, as the perceived value of their current program rises.

For outcomes measured at the application stage, the first cohort to fully observe a program's accredited status is the one applying and enrolling in year $t = 1$. This timing motivates our use of an event-study framework centered at $t = -1$, comparing outcomes over time relative to the year before accreditation is granted. Because programs are accredited at different points in time, we observe substantial staggered variation in treatment. This structure provides quasi-experimental variation in accreditation exposure across otherwise similar programs, enabling us to estimate dynamic treatment

[7] We explore alternative thresholds for inclusion and find consistent results, suggesting our findings are not driven by outlier programs.

[8] See Section 2 for further discussion of visibility and timing.

effects under standard identification assumptions.

Table 1 presents descriptive statistics for key program-level variables, comparing programs that eventually receive accreditation with those that never do in the pre-treatment period. The comparison reveals a compelling and complex story of selection, suggesting that the path leading to accreditation is not random.

On many dimensions of student demand and outcomes, programs that will eventually be treated appear stronger from the start. As shown in Panel A and B, they are larger and more popular programs that receive significantly more applications (71.9 vs. 53.8), attract academically stronger students (mean PSU score of 558.4 vs. 553.3), and enroll larger cohorts. They also appear to produce better outcomes, with lower first-year dropout rates and higher on-time graduation rates.

However, this straightforward story of quality is complicated by the financial and structural attributes shown in Panel C. Counter-intuitively, these seemingly more successful programs charge significantly lower tuition (approximately 0.25 million CLP less) and operate with a significantly lower faculty-to-student ratio (0.069 vs. 0.100). This presents a puzzle: why would programs that are more popular and produce better retention outcomes be cheaper and have fewer faculty resources per student?

This tension is central to the narrative of our paper. The combination of strong outcomes with fewer faculty resources suggests that these programs may be achieving their results through means other than deep investment. These systematic, and at times contradictory, differences underscore the endogeneity of accreditation and the impossibility of drawing causal conclusions from simple cross-sectional comparisons. This provides strong reasons to use an empirical approach that can control for these pre-existing heterogeneities and isolate the causal impact of the accreditation signal itself, such as our staggered difference-in-differences.

4 Empirical strategy

4.1 Eventy–Study Approach

To assess how students respond to a program's initial accreditation, we construct a panel dataset of undergraduate programs spanning 2007 to 2018. The dataset contains program-by-year aggregates of student behavior and characteristics, including application counts, enrollment figures, and measures of academic progression. A key component of the dataset is an indicator derived from CNA records that identifies the year in which each program receives accreditation for the first time.

Given that programs obtain accreditation at different points in time and retain this status thereafter, we employ a staggered adoption framework. This approach allows us to examine the dynamic impacts that follow a program's initial accreditation. We implement an event–study design, standardizing the event to occur at time $t = 0$, which corresponds to the calendar year in which a program p is first granted accreditation. Because accreditations are typically awarded during the year—often after the academic cycle has begun—students enrolled in $t = 0$ have already made their application and enrollment decisions. As such, $t = 1$ is the first cohort of students who can observe and respond to the program's newly accredited status. Nonetheless, $t = 0$ represents the timing of the treatment itself, when the program's status officially changes.[9] Our specification includes four pre-treatment periods (leads) and four post-treatment periods (lags) to capture both anticipatory dynamics and medium-run impacts. The number of lags is chosen to align with the average duration of accreditations granted, avoiding overlap with subsequent accreditation cycles. Our main specification is the following:

$$Y_{pt} = \mu_p + \lambda_t + \sum_{m=2}^{4} \delta_m (\text{Lead } m)_{pt} + \sum_{q=0}^{4} \gamma_q (\text{Lag } q)_{pt} + \Gamma X_{pt} + \varepsilon_{pt}, \tag{1}$$

where Y_{pt} captures the outcome of program p in year t, X_{pt} is a time–varying program-level control (depending on the outcome), μ_p and λ_t are program and year fixed effects, and ε_{pt} is an unobserved error term.

[9] Because accreditation is typically granted partway through the year, $t = 0$ captures the institutional timing of treatment, while $t = 1$ is the first year in which new students can observe and respond to the accreditation status.

Recent literature has highlighted that estimating equation (1), or its static counterpart—the difference-in-differences (DiD) specification—using the standard two-way fixed effects (TWFE) estimator in a staggered adoption setting can yield misleading results (Borusyak et al., 2024; Callaway and Sant'Anna, 2021; Goodman-Bacon, 2021; Roth et al., 2023; Sun and Abraham, 2021).

In the static case, TWFE estimators often rely on inappropriate comparisons, such as using early-treated programs as controls for later-treated ones, which can result in negative weighting and distorted treatment effect estimates. In the dynamic case, these issues are intensified when treatment effects vary across cohorts—for example, if the effect two years after accreditation differs between programs accredited in 2010 and those accredited in 2011. As shown by Sun and Abraham (2021), the resulting TWFE coefficients are linear combinations of cohort-specific effects from both relevant and irrelevant periods, leading to "contaminated" estimates that do not reflect any specific cohort's experience. Moreover, this contamination affects both post-treatment and pre-treatment periods, making even visual inspections of pre-trends potentially misleading.

To address these concerns, we implement the imputation estimator proposed by Borusyak et al. (2024), which avoids contamination and forbidden comparisons by estimating counterfactual outcomes for treated units based on not-yet-treated observations. This method is particularly well-suited to settings where a clean control group does not exist, as is the case for programs under mandatory accreditation, where all units eventually receive treatment. The estimator proceeds in two steps. First, it estimates the untreated potential outcomes by projecting the outcome variable onto fixed effects and control variables using only untreated observations. Second, for each event time, it imputes the counterfactual outcome for treated units using these estimated untreated outcomes and computes treatment effects as the difference between actual and imputed outcomes. This approach avoids extrapolation across cohorts and accommodates arbitrary treatment effect heterogeneity across units and over time.

Our identification strategy relies on the assumption that, absent accreditation, programs that become accredited and those that do not would have followed similar trends in the outcomes of interest. This parallel trends assumption is required both for conventional DiD estimators and for the imputation-based estimator we employ, which compares treated units to not-yet-treated ones. Importantly, the plausibility of this assumption depends on whether treatment timing is correlated with unobserved trends in program quality or student demand.

To assess this, we begin our analysis by examining program-level investment behavior among programs that voluntarily seek accreditation—such as changes in tuition levels, faculty composition, and infrastructure—before and after accreditation. These outcomes provide insight into whether programs systematically adjust their observable characteristics in anticipation of treatment. If we observe little to no change prior to accreditation, it suggests that accreditation reflects a program's existing attributes at the time of evaluation, rather than codifying a process of quality improvement. This supports the interpretation of accreditation as a credible signal that reveals new information to students, rather than one that merely formalizes prior investments.

Next, we study student application behavior, focusing on programs within the centralized admission system due to data limitations for those operating outside it. Using students' Rank-Order Lists, we estimate the effect of accreditation on both the volume and composition of first-preference applications received by programs. Specifically, we examine how accreditation shapes application patterns across different student subgroups, disaggregating responses by socioeconomic background—measured through income level and parental education—and by academic ability, defined through quartiles in college-entry scores.

We then analyze enrollment and persistence outcomes across the broader higher education system, including programs outside the centralized admission system. To capture changes in enrollment size, we use the log of the number of first-year students, which allows us to measure percentage changes in cohort size relative to the previous year. We then follow each first-year cohort into their second and third years to detect changes in attrition and transfer patterns. Persistence outcomes are defined as the share of a first-year cohort that is no longer enrolled at the same institution in subsequent years, distinguishing between those who permanently exit and those who transfer to programs at other institutions. In all specifications involving persistence, we include total first-year enrollment as a control to account for variation in cohort size across programs.

We interpret dynamic treatment effects relative to the year prior to accreditation ($t = -1$), allowing us to detect anticipatory responses, immediate shifts, and longer-run adjustments. We first focus on voluntary accreditation cases, where never-treated programs provide a credible comparison group. After establishing baseline patterns and exploring heterogeneity by institutional characteristics, we extend our analysis to mandatory accreditation programs, which require a separate estimation strategy due to the absence of untreated units. In these cases, we rely on the ability of the imputation estimator to recover treatment effects using only not-yet-treated programs as counterfactuals (Borusyak et al., 2024).

4.2 Difference-in-Differences Approach

Using the same set of outcomes described previously, we also estimate a static version of the treatment effect associated with a program's first accreditation. This provides a pooled estimate of the average difference in outcomes between treated and untreated programs, after accreditation has been granted. The specification is given by:

$$Y_{pt} = \mu_p + \lambda_t + \beta^{did} D_{pt} + \Gamma X_{pt} + \varepsilon_{pt}, \tag{2}$$

where D_{pt} is an indicator equal to 1 if program p has been accredited in year t or earlier, and 0 otherwise. The terms μ_p and λ_t are program and calendar year fixed effects, respectively. The coefficient β^{did} captures the average treatment effect of accreditation, pooling across all post-treatment years and assuming a constant effect across treated units and over time.

As discussed above, this specification imposes strong restrictions on treatment dynamics and homogeneity. Estimating equation (2) with a standard TWFE model can be problematic when adoption is staggered, due to contamination and negative weights. Therefore, we report estimates using the robust estimator proposed by Borusyak et al. (2024), which remains valid under heterogeneous treatment effects.

5 Student Responses to Accreditation

This section presents our main causal findings regarding student responses to first-time program accreditation. We analyze how the public announcement of an accreditation decision influences student application, enrollment, and persistence behavior, leveraging the plausibly exogenous timing of this information shock as discussed in Section 4.

5.1 Application Behavior

Column (1) of Table 4 presents the main estimate for first-choice applications. First-time accreditation leads to a statistically significant increase of 0.102 (s.e. 0.030) in the log number of applications. This corresponds to a 10.2% increase (approximately 6.4 applications) relative to the control group mean of 62.9 applications per program.

Accreditation also affects the composition of the applicant pool. Column (2) of Table 4 shows a statistically significant increase in the mean PSU score of applicants by 1.82 points (s.e. 0.972) relative to a control mean of 555.6 points. Furthermore, applications increase significantly for students with high prior academic achievement as log applications rise by 0.144 (s.e. 0.035) for students in the top score quartile, as shown in Column (3), a 14.4% increase (approximately 1.9 applications) relative to a control mean of 13.5 applications. For those above the 90th percentile, log applications increase by 0.088 (s.e. 0.029), as shown in Column (4), an 8.8% rise (approximately 0.5 applications) relative to a control mean of 5.2 applications.

Figures 2(a)-(d) present the corresponding event study plots for these application outcomes. Across all four measures, we find no statistically significant differences between treatment and control programs in the years preceding accreditation, with joint pre-trend p-values ranging from 0.229 to 0.794. These results support the parallel trends assumption underlying our identification strategy and indicate that students generally do not react in anticipation of the accreditation announcement. In each case, the estimated effects begin in year $t = 1$, consistent with the timeline through which accreditation becomes publicly visible to the next cohort of applicants.

We examine heterogeneity in application responses across socioeconomic backgrounds in Table 5. Point estimates for the increase in log applications are positive and statistically significant across all groups defined by household income, as shown in Columns (1) to (3), and parental education in Columns (4) to (6). For instance, log applications increase by 0.085 (s.e. 0.032) for students from low-income families, representing an 8.5% increase relative to a control mean of 25.4 applications for this group. Similarly, log applications increase by 0.078 (s.e. 0.032) for students whose parents have primary education, a 7.8% rise relative to a control mean of 8.7 applications. While point estimates vary slightly across groups, confidence intervals overlap substantially, and formal tests indicate no statistically significant differences in the treatment effect across these socioeconomic categories. The corresponding event study plots in Figures 3 and 4 visually confirm the similar positive response across groups starting at $t = 1$, with flat pre-trends.

Interpreting the dynamics behind the effects shown in Figures 2, 3, and 4 requires care. The coefficient at $t = 1$ represents the immediate effect of the information shock on the first cohort making application decisions with full knowledge of the new accreditation status. Coefficients for $t = 2$ through $t = 4$ reflect the persistence of this signal effect, potentially combined with equilibrium adjustments in the market or evolving student perceptions over time. The general stability of the positive effects across these lags suggests the signal retains its value for attracting students in the medium term.

5.2 Enrollment and Persistence

While application behavior reveals initial interest, enrollment and persistence outcomes show whether this interest translates into binding commitments and potentially improved student-program match.

Column (1) of Table 6 shows that first-time accreditation leads to a statistically significant increase in first-year enrollment. The estimated effect on log enrollment is 0.073 (s.e. 0.026), corresponding to a 7.3% increase (approximately 4.5 students) relative to the control mean of 61.3 students. The event study plot in Figure 5(a) confirms flat pre-trends and shows the increase beginning at $t = 1$, consistent with the timing of information release to applicants.

We also find effects on early-stage retention. Column (2) of Table 6 shows that first-time accreditation reduces the first-year attrition rate by 1.0 percentage point (-0.010, s.e. 0.005), relative to a baseline attrition rate of 23.2% in the control group. The event study (Figure 5(b)) shows this effect emerges in year $t = 0$, suggesting an immediate response among currently enrolled students.

Decomposing attrition over a three-year window reveals the margin of adjustment. Column (3) of Table 6 shows that transfers to other universities decline significantly by 1.0 percentage point (-0.010, s.e. 0.004), relative to a baseline transfer-out rate of 12.2%. In contrast, Column (4) illustrates that permanent exits from the higher education system show no statistically significant change (point estimate -0.003, s.e. 0.004) relative to a baseline rate of 6.9%. Figures 5(c) and (d) display the event study dynamics, confirming flat pre-trends for both outcomes.

The dynamic patterns of the effects on enrollment and persistence, shown in Figure 5, also require careful interpretation. The enrollment effect mirrors the application effect, appearing at $t = 1$. For persistence outcomes such as attrition and transfers, the effects emerge at $t = 0$ or $t = 1$ and remain relatively stable through $t = 4$. The coefficients in these later periods ($t \geq 2$) may reflect not only the enduring effect of the signal but also changes in cohort composition, as the academically stronger students attracted by accreditation progress through the program.

6 Heterogeneous Responses to Accreditation

The average effects of accreditation presented in Section 5 may mask important variation in how the signal is interpreted and acted upon across different settings. We now explore how the impact on student outcomes is mediated by two key contextual factors: pre-existing institutional reputation[10] and the regulatory framework under which program accreditation is granted.

6.1 Institutional Reputation

The value of program-level accreditation may depend on the broader reputation of its host university. Economic theory suggests formal certification and informal reputation can act as either substitutes or complements (Börgers et al., 2013; Colombo et al., 2019; Elfenbein et al., 2012, 2015). If institutional prestige already provides a strong signal, program-level accreditation might have limited marginal value (substitutes). Conversely, if the signals provide distinct information or reinforce each other, accreditation's effect could be stronger at more reputable institutions (complements). To test these competing hypotheses, we estimate a fully specified interaction model based on equation (2), incorporating indicators for different tiers of institutional quality.

A critical methodological point arises when moving from the aggregate to the subgroup analysis. While the main student response results satisfy the parallel trends assumption, as showed in Section 5, the disaggregated event studies in Appendix Figures A.12-A.15 reveal that this assumption does not

[10] We use the length of institutional accreditation granted by the CNA as a proxy for institutional quality/reputation. The CNA is the only agency authorized to accredit higher education institutions in Chile, and over time its accreditation decisions have created an implicit ranking (Barroilhet, 2019). Alternative measures, like selectiveness (Bordón et al., 2020), yield consistent results.

hold for all outcomes within specific institutional tiers. Notably, enrollment and first-year attrition exhibit clear pre-trends for certain subgroups, making impossible the causal interpretation of their dynamic patterns. This finding does not invalidate the average treatment effects estimated on the full sample but requires careful interpretation of heterogeneous effects. Instead, these differential pre-trends are themselves informative, suggesting that programs at different quality tiers may seek accreditation under different circumstances, perhaps as a strategy to consolidate market share or as a remedial tool (Cameron et al., 2023). Consequently, our causal analysis of heterogeneous effects by reputation is restricted to the subgroups and outcomes where the parallel trends assumption holds, like transfers-out and permanent exits.

Further context for these differential pre-trends comes from the distribution of accreditation outcomes themselves. Figure 8 displays the share of first-time accreditations receiving each possible duration (1-7 years), separated by institutional quality tier. The figure shows a distinct pattern, as accreditations granted to programs at baseline and enhanced institutions are more concentrated at shorter durations (e.g., 2-3 years), whereas top-tier institutions exhibit a wider distribution with a greater share receiving longer accreditations (e.g., 5-7 years). This descriptive correlation between institutional tier, pre-treatment enrollment trends, and the final accreditation outcome provides suggestive evidence consistent with the idea that institutions may select into accreditation under different circumstances, potentially related to underlying quality trajectories or strategic goals. However, as noted, the presence of pre-trends limits causal interpretation for some outcomes within these subgroups.

Table 7 presents the estimated heterogeneous effects for persistence outcomes where pre-trends are absent. For transfer-out rates within three years, shown in Column (3), programs at baseline institutions experience a statistically significant decrease of 1.1 percentage points (-0.011, s.e. 0.005) post-accreditation, representing a substantial 14.7% reduction relative to the baseline transfer rate of 7.5% for this group. In contrast, the estimate for top-tier institutions is small and statistically insignificant (-0.003, s.e. 0.005) compared to their baseline rate of 19.4%. Permanent exits within three years, in Column (4), also show differential effects: programs at baseline institutions exhibit a statistically significant increase of 1.4 percentage points (0.014, s.e. 0.005), corresponding to a 14.6% increase relative to their baseline exit rate of 9.6%. Conversely, programs at enhanced and top-tier institutions show statistically significant reductions of 1.2 percentage points (-0.012, s.e. 0.001) and 1.5 percentage points (-0.015, s.e. 0.002), respectively. These represent sizable relative decreases compared to their baseline rates of 5.5% for programs at mid-tier (Enhanced) institutions and 4.4% for programs at top-tier institutions.

6.2 Regulatory Framework

We complement the analysis by examining heterogeneity based on the regulatory framework. In contrast to voluntary accreditation, where self-selection into the process is potentially informative (Spence, 1973), mandatory regimes eliminate this margin (Dranove and Jin, 2010; Jin, 2005). This analysis adds 304 programs from fields where accreditation is mandatory (Medicine, Odontology, Pedagogy), using not-yet-accredited programs in those same fields as controls. We estimate a fully specified interaction model based on equation (2), including an indicator for the regulatory framework.

Appendix Figures A.16-A.19 confirm parallel pre-trends for the student outcomes analyzed in this subsection, supporting the validity of comparing effects across regimes. Table 8 presents the results. The estimates for voluntary programs replicate those reported in Section 5. Focusing on the mandated programs, we find a statistically significant reduction in log first-year enrollment in Column (1), of 0.201 (s.e. 0.017) following accreditation. This corresponds to a large 18.2% decrease relative to the average enrollment size, or approximately 10 fewer students compared to the control mean of 56.1 students in mandated programs. Simultaneously, mandated programs exhibit statistically significant reductions across several attrition measures. First-year dropout, exhibited in Column (2), decreases by 1.2 percentage points (-0.012, s.e. 0.004), a 6.6% reduction relative to the baseline rate of 18.3%. Transfers-out within three years, in Column (3), decrease by 0.9 percentage points (-0.009, s.e. 0.002), a 13.4% reduction relative to the baseline rate of 6.7%. Permanent exits within three years, Column (4), decrease by 0.3 percentage points (-0.003, s.e. 0.003), a 4.6% reduction relative to the baseline rate of 6.5%.

The distribution of accreditation outcomes offers context for these patterns, particularly the enrollment decline under the mandatory framework. Figure 9 shows the share of first-time accreditations receiving each duration ($1-7$ years) under both regimes. Programs accredited under the mandatory regime are significantly more likely to receive shorter accreditation periods (e.g., $2-3$ years) compared to those in the voluntary regime, which show a wider distribution skewed towards longer durations. This difference in outcomes suggests that the informational content or interpretation of the signal might vary significantly between the two frameworks. While the mere status of being accredited might carry less weight when it is mandatory, the awarded duration could potentially become a more salient indicator for students.

7 Institutional Dynamics and Selection into Accreditation

Having established that students respond significantly to the accreditation signal in Section 5, this section turns to our third research question: what is the nature of the signal students are responding to? We provide a descriptive analysis of institutional behavior in the years surrounding a program's first accreditation event. Specifically, we examine whether programs undertake observable long-term, structural investments or focus on adjustments to more malleable metrics that align closely with the evaluation cycle, as detailed in Appendix A. Understanding these pre-accreditation dynamics provides crucial context for interpreting the signal's content.

7.1 Institutional Adjustments Around First-Time Accreditation

We first examine whether programs undertake significant long-term, structural investments such as expansions in infrastructure, faculty, or declared capacity in the years preceding their first accreditation. Observing such investments would suggest the signal reflects recent improvements, while their absence would indicate the signal primarily certifies existing program attributes at the time of evaluation.

Figures 6(a)-(d) report event-study estimates for key structural inputs: declared enrollment capacity, buildings per student, campus size per student, and faculty-to-student ratio. Across all four measures, we observe no statistically significant pattern of anticipatory adjustments in the years leading up to accreditation relative to control programs. F-tests of joint significance for the pre-treatment periods fail to reject the null hypothesis of parallel trends (p-values range from 0.349 to 0.661). Post-treatment dynamics are similarly flat, and the static DID estimates reported Columns (1) through (4) in Table 2 are small and statistically insignificant. Even for lecturer availability, arguably a more flexible input, we find no evidence of differential changes prior to accreditation (ATT $= -0.002$, s.e. 0.003). This lack of observable pre-accreditation investment in structural inputs contrasts with the adjustments observed in other metrics. This finding aligns with signaling models where agents may forgo costly, hard-to-observe investments in fundamental quality if lower-cost actions can achieve the desired signal (Dranove and Jin, 2010; Jin, 2005; Spence, 1973).

In contrast to structural inputs, we observe pre-treatment dynamics in more malleable program metrics. Figure 7(a) shows event-study estimates for annual tuition. We observe a gradual increase in tuition, accumulating to approximately 1% over the three years prior to accreditation ($t = -3$ to $t = -1$). Post-accreditation, tuition tends to decline relative to the control group. One possible interpretation of the pre-treatment rise is that programs increase prices temporarily, perhaps to signal quality improvements or offset evaluation costs, before prices revert post-accreditation.

A more pronounced pattern emerges for on-time graduation rates, shown in Figure 7(b).[11] The event study reveals a rising trend throughout the pre-treatment period, culminating in a statistically significant positive deviation at $t = -1$. Following accreditation, however, the estimates suggest this upward trend flattens and does not persist relative to control programs. This temporal pattern (improvement concentrated before evaluation and not sustained after) is consistent with theoretical predictions about strategic behavior in disclosure markets, where firms may focus effort on evaluated metrics around the assessment period (Dranove and Jin, 2010; Jin, 2005). Given that on-time graduation is an explicit input in the CNA's accreditation assessments (Appendix A), this pattern suggests programs may allocate effort towards this metric specifically during the accreditation cycle.

Alternative interpretations exist. For instance, the pre-accreditation rise could reflect improvements genuinely valued by students, such as smoother administrative processes facilitating timely graduation. However, the specific timing alignment of our measure, which links graduation year to the accreditation cycle, which often happens years after a cohort initially enrolled, makes it seem more likely that the pattern is driven by program-level efforts rather than solely by changes in underlying

[11] Recall from Section 3 that this outcome is aligned relative to a cohort's expected graduation year, unlike other outcomes aligned to the enrollment year.

instructional quality or student-driven acceleration. Another possibility is that even if the adjustments are short-term, they might function as a costly signal of the institution's underlying capabilities or commitment to meeting standards, which could be how students interpret the subsequent positive accreditation outcome. Nonetheless, the clear concentration of the increase immediately before evaluation, together with the lack of persistence, strongly suggests that the adjustments are timed specifically to the certification process. In the Chilean context, such short-term increases can potentially be achieved through administrative efforts like outreach to near-completers or deadline flexibility, without necessarily reflecting underlying long-term improvements in instructional quality.

Table 2 summarizes the static estimates and pre-trend tests for all institutional response variables. The results quantitatively highlight the contrast observed in the event studies, as statistically insignificant pre-trends and average effects for structural inputs shown in Columns (1) to (4), but statistically significant pre-trends for tuition and on-time graduation rates found in Columns (5) and (6).

These adjustment patterns are not uniform across institutions. Disaggregating by institutional quality reveals some heterogeneity (Table 3).[12] For instance, the pre-accreditation increase in tuition appears most pronounced among programs at enhanced institutions. Similarly, the pre-treatment rise in on-time graduation rates is evident for programs at baseline and enhanced institutions (Appendix Figures A.11 and A.10). We also observe a slight positive pre-trend in staffing for programs at baseline institutions. These differential patterns suggest that the incentives or ability to adjust specific metrics prior to accreditation may vary with institutional standing. This is particularly relevant given that accreditation often functions as a signal to distinguish programs from the "left tail" of the quality distribution (Akerlof, 1970); the incentives to manage observable metrics might be strongest for institutions competing just above a perceived minimum quality threshold. These dynamics could relate to factors such as market position or peer comparisons, as explored in related literature (Cameron et al., 2023).

Overall, the descriptive analysis indicates that while programs seeking accreditation do not show evidence of increasing long-term structural inputs, they do exhibit adjustments in more malleable metrics, particularly on-time graduation rates, timed closely with the evaluation cycle. It is important to

[12]We proxy institutional quality by the length of accreditation provided by the CNA. The public agency is the only one authorized to accredit higher education institutions in Chile. Over time, its accreditation decisions have created an implicit ranking of institutional quality (Barroilhet, 2019).

emphasize that these observed adjustments relate to metrics explicitly considered in the accreditation assessment (Appendix A) and do not necessarily reflect changes in the underlying, unobserved human capital formation process or the inherent quality of the educational experience. Understanding these patterns primarily provides context for the nature of the signal itself, as the information to which students demonstrably respond.

8 Conclusion

This paper examines the effects of program-level accreditation in Chilean higher education, analyzing both student responses and institutional adjustments surrounding the signal. Using a difference-in-differences approach exploiting the staggered adoption of first-time accreditation, we find that the signal significantly increases student demand: first-choice applications rise by 10.2% and enrollment by 7.3%, attracting academically stronger students. Furthermore, accreditation improves student-program match, primarily by reducing transfers to other institutions. These positive student responses are consistent across socioeconomic backgrounds, suggesting the signal is broadly accessible. Our heterogeneity analysis reveals variation based on context, as the effects differ by institutional reputation, consistent with complementarity where pre-trends allow causal interpretation, and differ markedly under mandatory regulatory regimes. Descriptively analyzing institutional behavior, we find no evidence of increased long-term structural investments prior to accreditation. Instead, we document adjustments in malleable metrics, notably a rise in on-time graduation rates concentrated immediately before evaluation and not sustained afterward.

Synthesizing these findings reveals a central insight: students respond strongly to the accreditation signal, enhancing market sorting, even though the institutional behavior preceding the signal appears focused on improving specific, evaluated metrics timed to the review cycle rather than broader structural investments. This aligns with theories of disclosure where firms may strategically manage observable information when deeper quality changes are costly or difficult to signal. This points to a crucial policy trade-off. From an allocative efficiency perspective, the signal appears beneficial, guiding students—including those from less-advantaged backgrounds—towards programs where they persist longer. However, from a productive efficiency standpoint, the evidence raises questions. If institutions substitute effort towards evaluated metrics for potentially more impactful long-term investments in human capital formation, the policy may improve matching without necessarily increasing overall quality or productivity.

The heterogeneous effects provide further nuance and potential explanations for observed behaviors. Crucially, the differing pre-treatment enrollment trends across institutional tiers suggest selection effects related to the motivation for seeking accreditation. Baseline and enhanced institutions, which exhibit potentially concerning pre-trends and also tend to receive shorter accreditation durations as exhibited in Figure 8, might seek accreditation more often as a remedial tool or signal (Cameron et al., 2023). The subsequent persistence patterns for baseline institutions, including reduced transfers but increased permanent exits, could reflect this as a weak positive signal might be enough to retain students considering lateral moves ("better than no signal at all"), but might also coincide with increased academic rigor that pushes marginal students out entirely. For outcomes without pre-trends, the concentration of reductions in permanent exits among mid- and top-tier institutions hints at complementarity between accreditation and reputation, suggesting formal certification reinforces strong existing signals (Börgers et al., 2013; Colombo et al., 2019). The analysis by regulatory framework reveals starkly different patterns. The enrollment decline under mandatory accreditation contrasts sharply with the positive effect under voluntary regimes. Given that mandatory status removes informative self-selection (Dranove and Jin, 2010; Jin, 2005) and mandatory programs often receive shorter accreditation durations as Figure 9 indicates, students might interpret short mandatory accreditations as a negative signal relative to expectations, leading to reduced demand. However, for enrolled students or those who enter regardless, the observed decrease in subsequent attrition and transfers suggests a different interpretation, that even a short mandatory accreditation might signal sufficient relative quality compared to other programs that must also eventually seek certification.

Our descriptive analysis of institutional behavior illuminates the nature of the signal students respond to. The lack of observable pre-accreditation investment in long-term structural inputs, coupled with the temporal pattern observed for on-time graduation rates—a rise concentrated before evaluation and not sustained afterward, is consistent with theories of strategic disclosure and signal management (Dranove and Jin, 2010; Jin, 2005). When certification relies on observable metrics, especially those explicitly evaluated by the certifying body (Appendix A), firms face incentives to focus effort on those margins around the evaluation period. The concentration of these adjustment patterns among baseline and enhanced institutions aligns with the idea that accreditation serves primarily as a signal to differentiate from the "left tail" (Akerlof, 1970); incentives to manage observable metrics may be strongest for programs competing above a minimum quality threshold. It is crucial, however, to interpret these findings cautiously: they reflect adjustments to evaluated metrics and do not necessarily capture changes in unobserved aspects of educational quality or human capital formation.

These findings contribute to several literatures. First, by examining both the institutional adjustments preceding a signal and the market's response, we offer a more complete view of **disclosure markets**, extending empirical work that often takes the signal's content as given (Dranove and Jin, 2010; Jin and Leslie, 2003). Second, we add to the literature on information in higher education by providing large-scale causal evidence on how students react to an existing, institutionalized signal, demonstrating its effectiveness across socioeconomic groups (Dynarski et al., 2021). Third, our findings on heterogeneity by reputation provide evidence consistent with complementarity between formal accreditation and informal reputation in this context, adding to the literature on signal interaction (Bar-Isaac and Tadelis, 2008; Elfenbein et al., 2012; MacLeod et al., 2017; MacLeod and Urquiola, 2015).

Our analysis has limitations. As detailed in Appendix A.6, our measures of institutional investment are quantitative and may miss changes in input quality or investments in unobserved areas. Furthermore, while our identification strategy leverages the information shock to students, the endogeneity of the application timing remains a challenge for interpreting pre-treatment institutional trends. These limitations open avenues for future research. While accreditation appears to improve the initial student-program match, tracing cohorts into the workforce is crucial to determine if these benefits translate into improved long-run labor market outcomes, which is the ultimate measure for a complete welfare analysis. In future work we plan to leverage newly acquired administrative data on graduate earnings and employment to address this question directly. Additionally, exploring the industrial organization of the accreditation market itself, particularly the role of private agencies during our study period, represents a promising direction.

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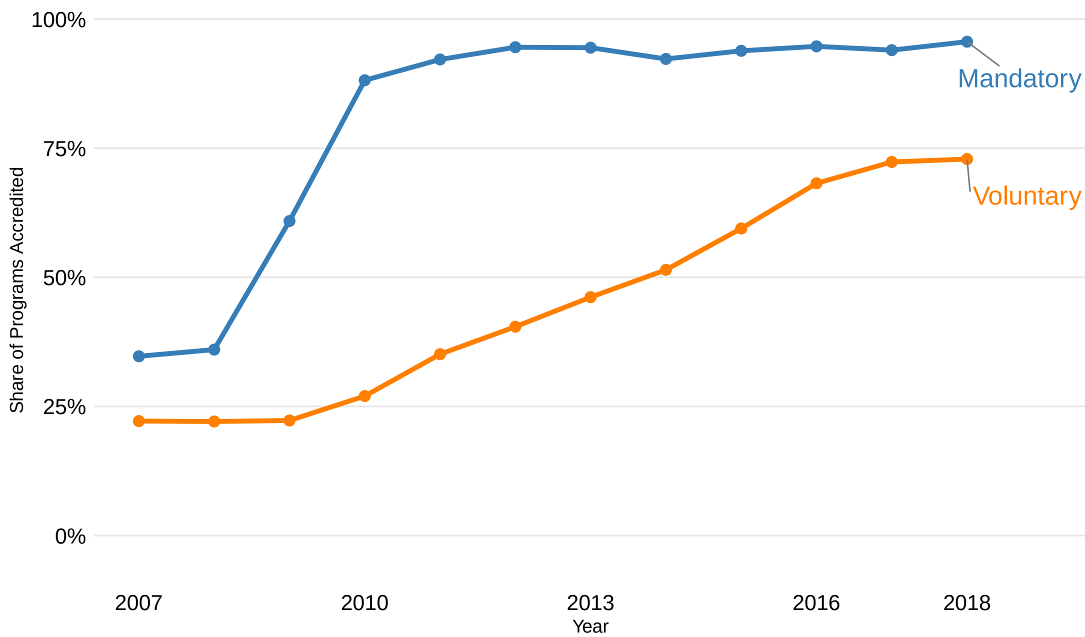
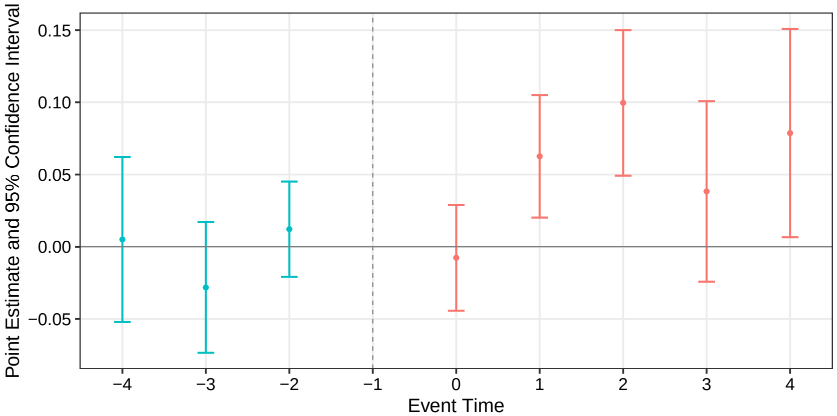
Figure 1: Accreditation Share of Accredited Programs by Regime



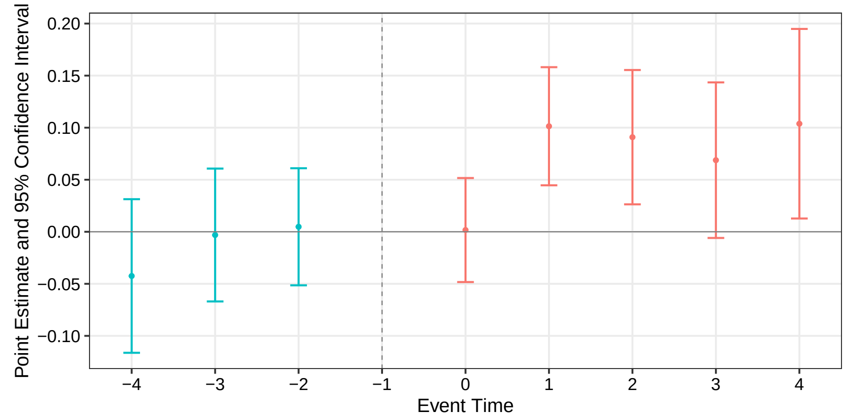
Note: This figure displays the cumulative adoption rate of accreditation for programs under both mandatory and voluntary regimes. Programs under the mandatory framework achieved near-universal accreditation shortly after the system's officialization. In contrast, the voluntary regime shows a gradual but steady adoption, with the share of accredited programs rising from approximately 25% to 75% over the decade.

Event study plots: Students Application Behavior

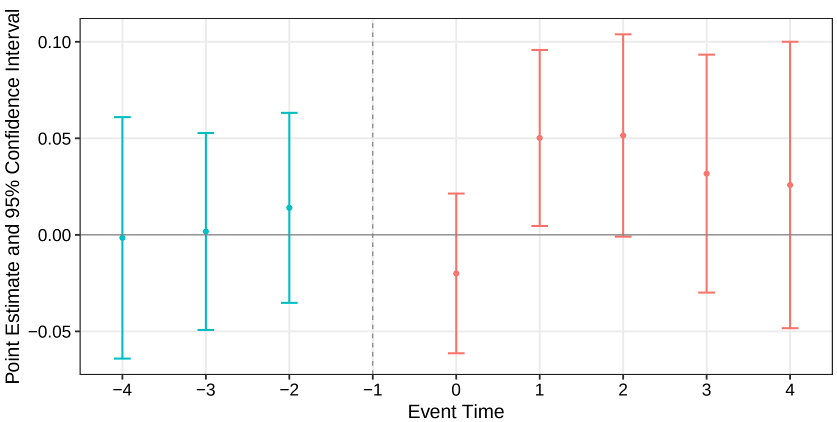
Figure 2: Event study estimates: Students application behavior



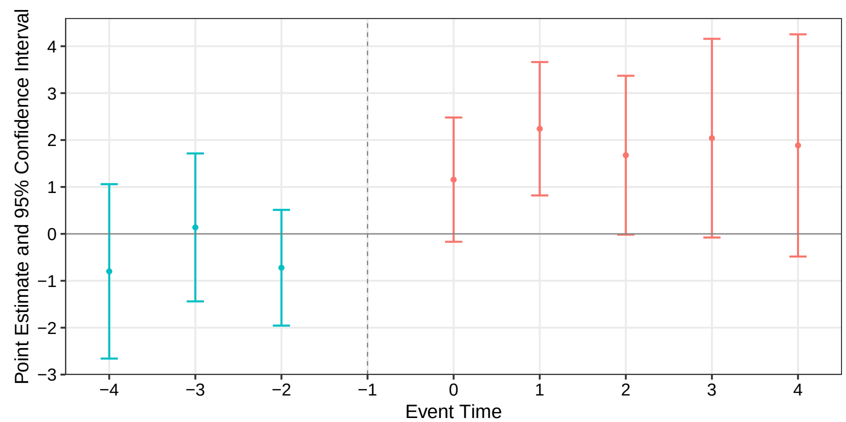
(a) Total applications



(b) Applications: top 75% scores



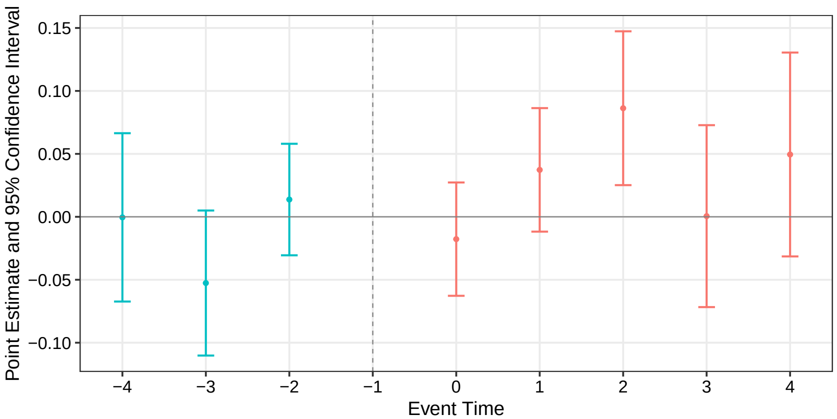
(c) Applications: top 90% scores



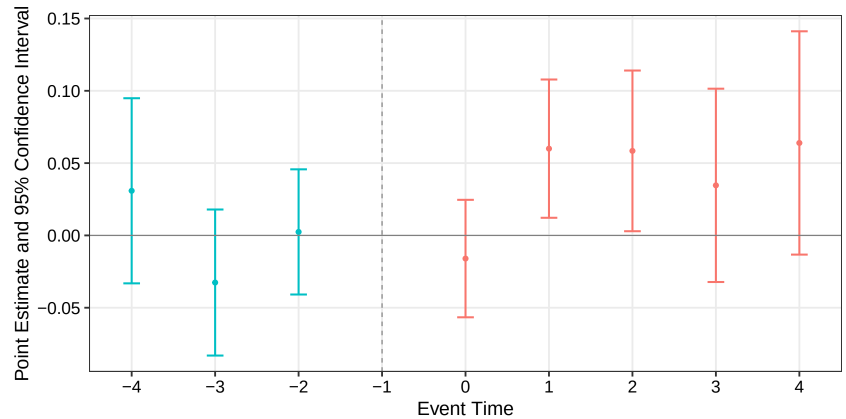
(d) Applications: mean score

Note: This figure plots event study estimates for the effect of first-time accreditation on student application behavior: (a) total applications, (b) applications from the top 75th percentile of scorers, (c) applications from the top 90th percentile of scorers, and (d) mean applicant score. The year prior to treatment ($t-1$) is the omitted reference category. Vertical bars represent 95% confidence intervals. Estimates follow Borusyak et al. (2024) with standard errors clustered at the program level.

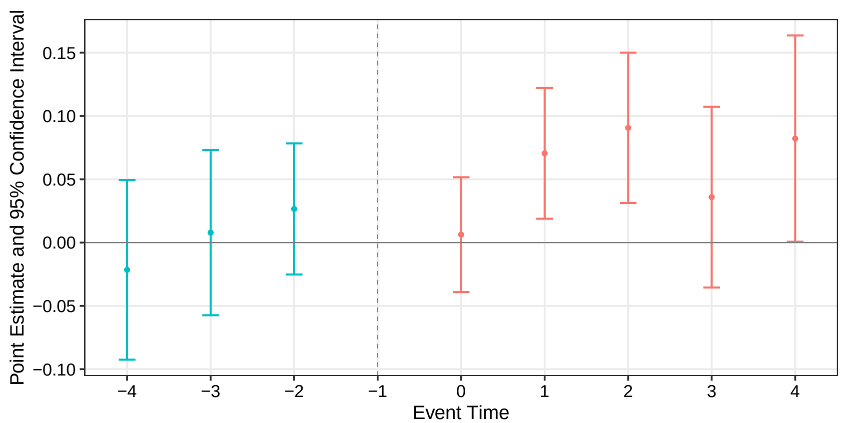
Figure 3: Event study estimates: Application behavior by parental income



(a) Low-income families



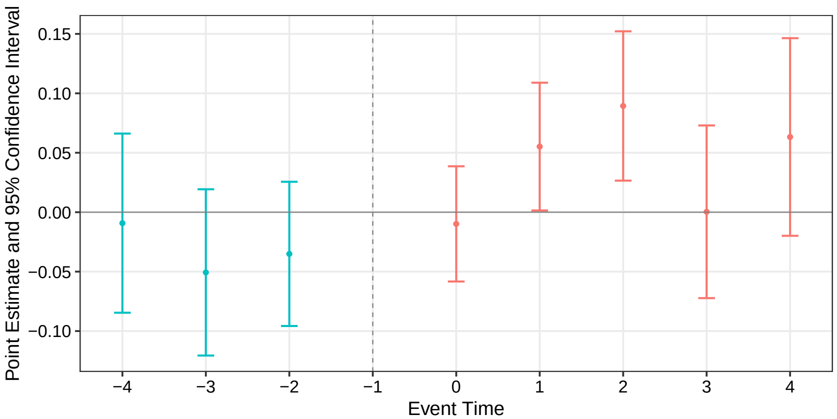
(b) Mid-income families



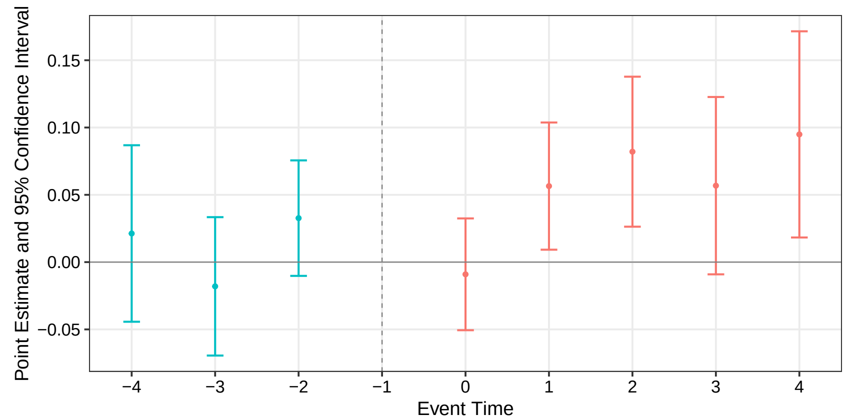
(c) High-income families

Note: This figure plots event study estimates for the effect of first-time accreditation on application behavior, disaggregated by parental income: (a) low-income families, (b) mid-income families, and (c) high-income families. The year prior to treatment ($t-1$) is the omitted reference category. Vertical bars represent 95% confidence intervals. Estimates follow Borusyak et al. (2024) with standard errors clustered at the program level.

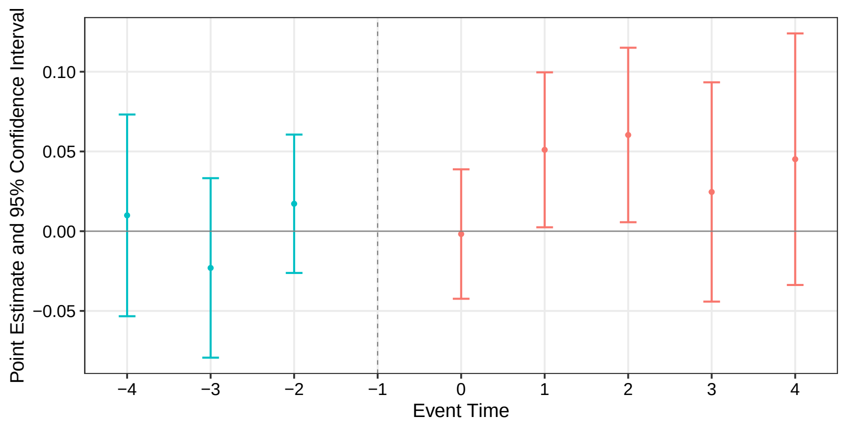
Figure 4: Event study estimates: Applications behavior by parental education



(a) Primary education



(b) Secondary education

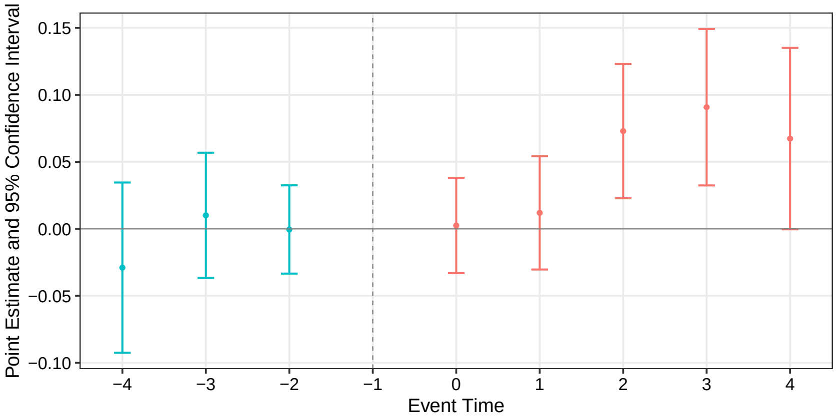


(c) College degree

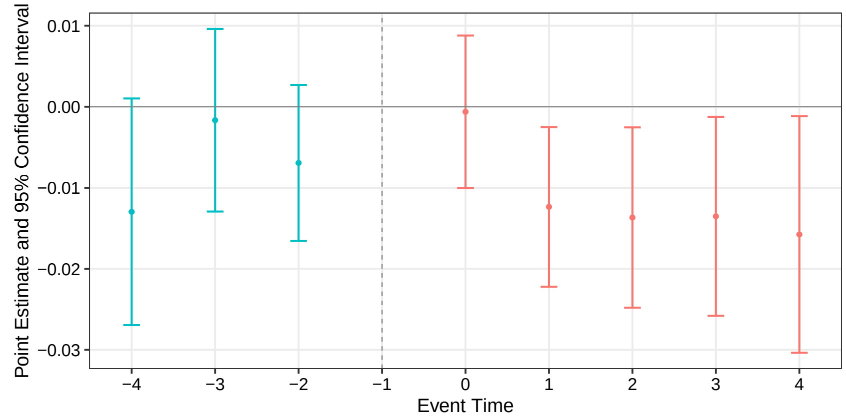
Note: This figure plots event study estimates for the effect of first-time accreditation on application behavior, disaggregated by parental education: (a) primary education, (b) secondary education, and (c) college degree. The year prior to treatment ($t-1$) is the omitted reference category. Vertical bars represent 95% confidence intervals. Estimates follow Borusyak et al. (2024) with standard errors clustered at the program level.

Event study plots: Students Enrollment and Persistence Behavior

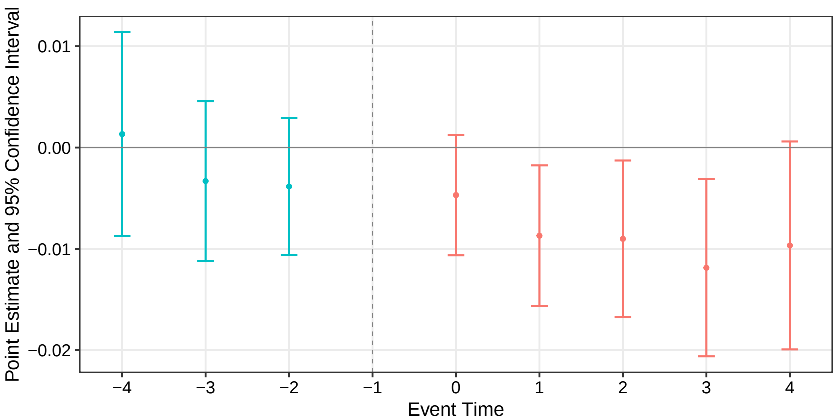
Figure 5: Event study estimates: Student enrollment and persistence behavior



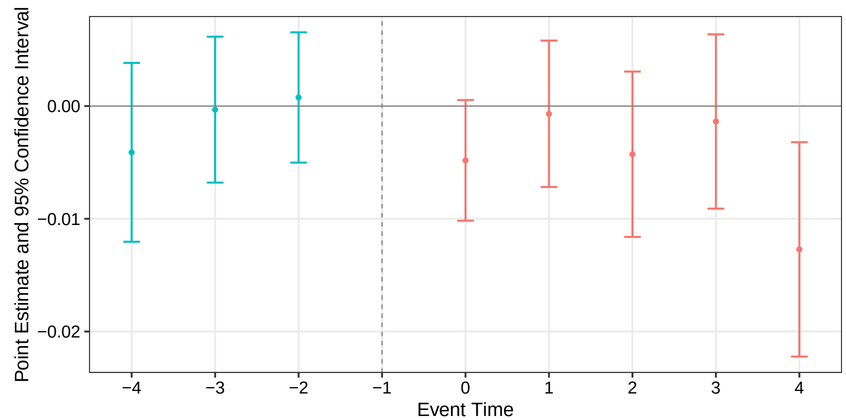
(a) First-year cohort



(b) First-year dropout



(c) Third-year transfers out

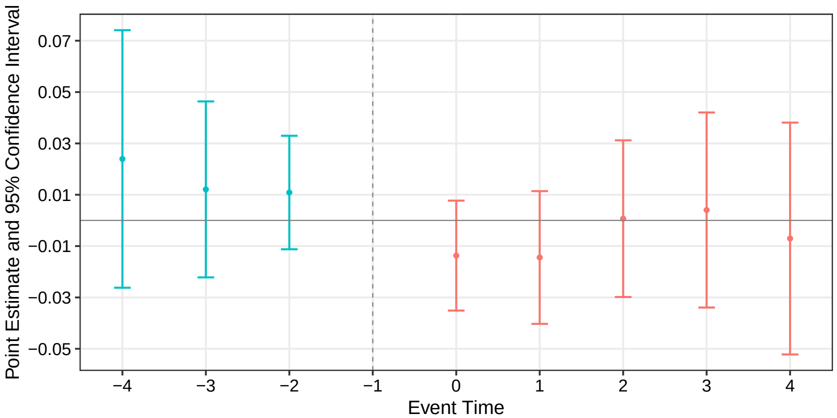


(d) Third-year exits

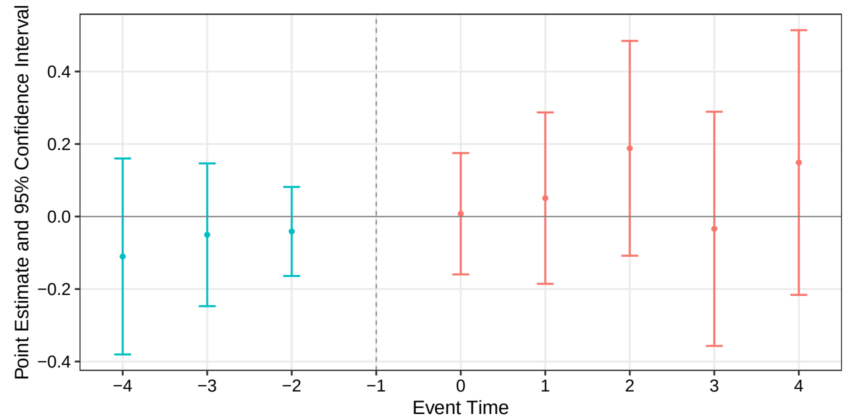
Note: This figure plots event study estimates for the effect of first-time accreditation on student enrollment and persistence: (a) first-year cohort size, (b) first-year dropout rate, (c) third-year transfers to other institutions, and (d) third-year program exits. The year prior to treatment $(t-1)$ is the omitted reference category. Vertical bars represent 95% confidence intervals. Estimates follow Borusyak et al. (2024) with standard errors clustered at the program level.

Event study plots: Programs Investment Outcomes

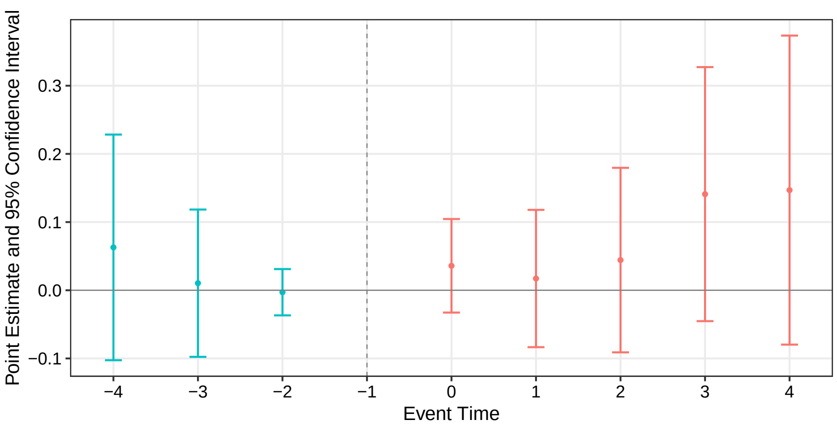
Figure 6: Event study estimates: Major investment outcomes



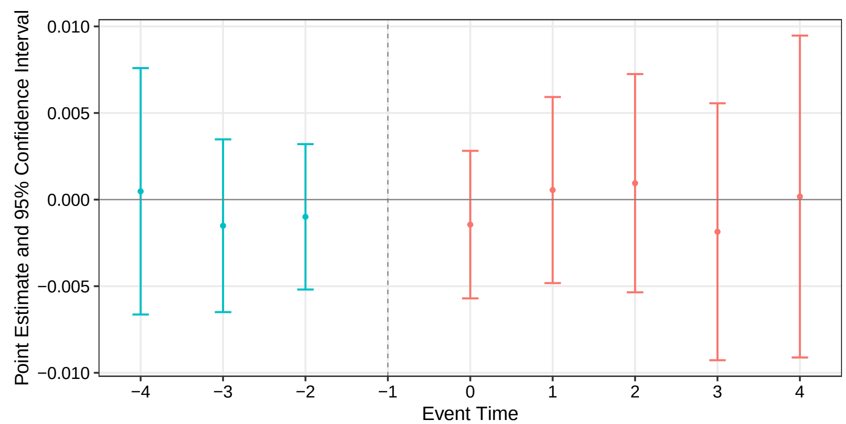
(a) Declared capacity



(b) Buildings (per 1000 students)



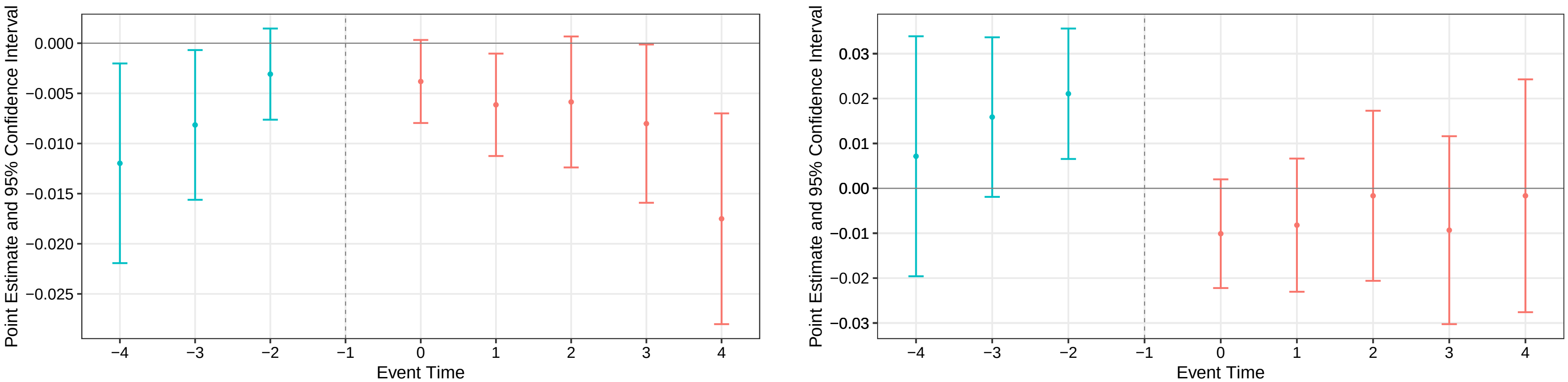
(c) Campus size (mt^2 per student)



(d) Lecturers per student

Note: This figure displays event study estimates of first-time accreditation on four measures of institutional investment: (a) declared capacity, (b) buildings, (c) campus size, and (d) lecturers. The horizontal axis represents years relative to the accreditation event, with the year prior to treatment ($t-1$) serving as the omitted reference category. Point estimates are shown with vertical bars representing 95% confidence intervals. Standard errors are clustered at the program level, and estimates are derived using the method from Borusyak et al. (2024).

Figure 7: Event study estimates: Strategic Metrics

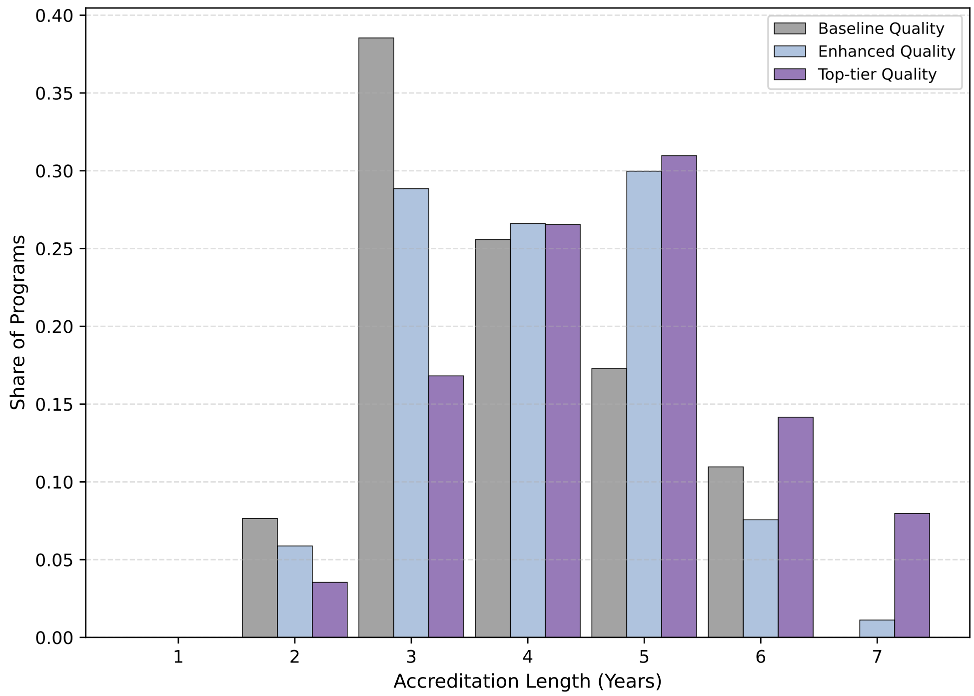


(a) Tuition

(b) On-time graduation

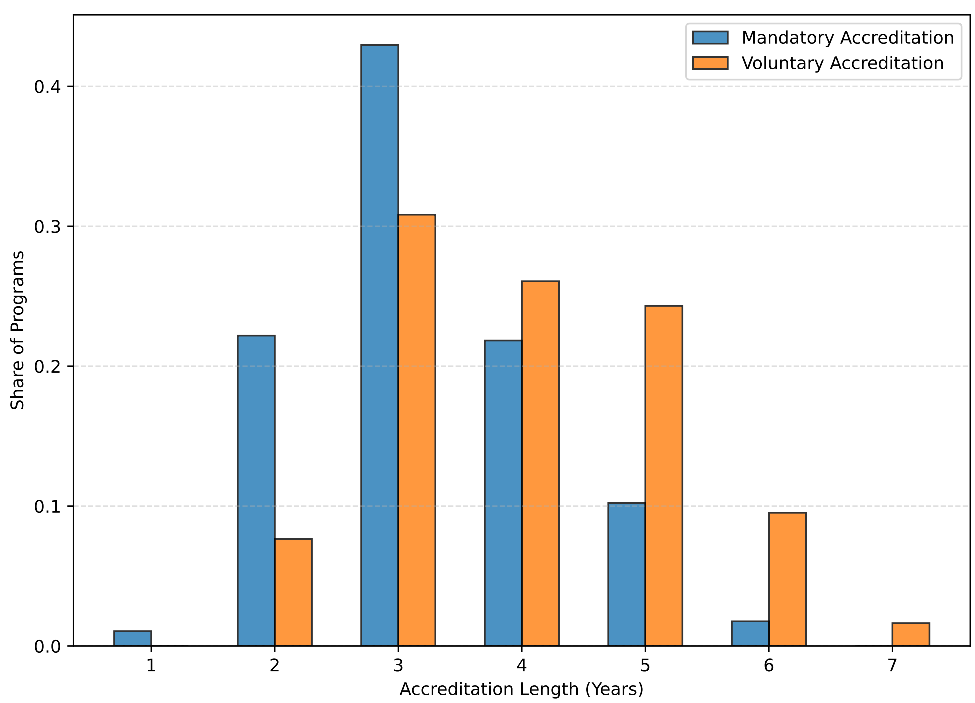
Note: This figure displays event study estimates of first-time accreditation on two strategic metrics: (a) Tuition and (b) On-time graduation rates. The horizontal axis represents years relative to the accreditation event, with the year prior to treatment $(t-1)$ serving as the omitted reference category. Point estimates are shown with vertical bars representing 95% confidence intervals. Standard errors are clustered at the program level, and estimates are derived using the method from Borusyak et al. (2024).

Figure 8: Distribution of Accreditation Length by Institutional Quality



Note: This figure plots the distribution of accreditation lengths, in years, disaggregated by three tiers of institutional quality. Institutional quality is defined based on the institutional accreditation.

Figure 9: Distribution of Accreditation Length by Regime



Note: This figure plots the distribution of accreditation lengths, in years, disaggregated by regulatory framework. It contrasts the outcomes for programs subject to the mandatory regime with those participating in the voluntary regime.

Table 1: Descriptive Statistics

	Control	Treated	Difference
Panel A: Applicant Pool Characteristics			
Number of Applications	53.785	71.934	18.149***
	(86.303)	(82.519)	(2.161)
Applicant Mean Score	553.288	558.389	5.100***
	(39.245)	(39.430)	(1.004)
Applications from Low-Income	19.999	29.879	9.880***
	(26.874)	(28.759)	(0.708)
Applicants College-Ed. Parents	20.396	27.520	7.124***
	(45.414)	(46.526)	(1.172)
Panel B: Enrolled Cohort Outcomes			
Number of Enrolled Students	47.601	68.854	21.253***
	(40.186)	(67.114)	(1.403)
First-Year Dropout Rate	0.251	0.219	-0.032***
	(0.147)	(0.122)	(0.003)
On-Time Graduation Rate	0.184	0.225	0.041***
	(0.180)	(0.190)	(0.005)
Panel C: Financial and Structural Attributes			
Tuition (CLP, thousands)	2816.624	2559.984	-256.639***
	(749.454)	(736.920)	(21.575)
Faculty-to-Student ratio	0.100	0.069	-0.031***
	(0.152)	(0.086)	(0.003)
N (Programs)	317	729	-

Note: This table displays the means of program-level characteristics for the Control and Treated groups. Standard deviations are reported in parentheses beneath the means. The unit of observation is the academic program. The "Difference" column reports the coefficient from an OLS regression of the variable on a treatment status indicator. The N (Programs) row shows the number of unique programs in each group. Significance levels are denoted as follows: * p<0.1, ** p<0.05, *** p<0.01.

Table 2: Estimated Effects of Accreditation on Programs response

	Major Investments				Strategic Metrics	
	Slots	Number of buildings	Size of campus	Lecturers	Tuition	On-time grads
ATT (Borusyak, Jaravel, Spiess)	-0.00763	0.00002	-10.99274	-0.00411	-0.00966**	0.00377
	(0.01788)	(0.00015)	(15.36979)	(0.00289)	(0.00446)	(0.00884)
Num. Obs	9021	10142	10142	9801	9056	11476
Control outcome mean	58.563	0.004	49.037	0.104	2684719	0.213
P-value pre-trends	0.451	0.661	0.581	0.349	0.0375	0.0753

Note: Standard errors (clustered at the program level) are in parentheses. The control outcome mean represents the average number of students in the control group for each outcome. All programs are offered by accredited institutions. The treatment group includes programs accredited for the first time. Average treatment effects on the treated follow Borusyak et al. (2024). * p<0.1, ** p<0.05, *** p<0.01.

Table 3: Estimated Effects on Programs response by Institutional Quality

	Major Investments				Strategic Metrics	
	Slots	Number of buildings	Size of campus	Lecturers	Tuition	On-time grads
Panel A: Implied Total Effects by Institutional Quality						
Baseline (β_1)	-0.08931***	-	-49.07977***	-	0.00867**	-
	(0.02648)	-	(15.32656)	-	(0.00378)	-
Enchanced (β_1 + β_2)	0.01154	0.00014	-48.34325***	-0.00111	-	-
	(0.00940)	(0.00004)	(1.49199)	(0.00152)	-	-
Top-tier (β_1 + β_3)	0.12648***	-0.00093***	165.02239***	-0.01680***	-0.00953***	-0.02964
	(0.01333)	(0.00018)	(32.49223)	(0.00172)	(0.00145)	(0.00960)
Num. Obs	7340	6671	9223	6446	2693	1427
Control Mean						
Baseline	53.12	-	13.045	-	1.046	-
Enhanced	61.659	0.002	26.527	0.082	-	-
Toptier	54.776	0.01	156.149	0.106	1.042	0.204
P-value Pre-trends						
Baseline	0.241	0.059	0.507	0.054	0.828	0.033
Enhanced	0.609	0.187	0.323	0.174	0.003	0.04
Top-tier	0.899	0.11	0.323	0.537	0.644	0.611

Note: First panel reports the implied effects of accreditation for each institutional category (Basic, Advanced, and Excellence). Standard errors are clustered at the program level. Control outcome means reflect the average in the control group by category. Accreditation levels follow CNA guidelines: Basic (reference), Advanced, and Excellence. Omitted coefficients are due to evidence of differential pre-trends for that institutional category, violating the parallel trends assumption, following the p-values for the F tests on pre-treands. Estimates follow Borusyak et al. (2024). * p<0.1, ** p<0.05, *** p<0.01.

Table 4: Estimates of receiving accreditation on applications levels

	First-Choice Applications	Applicant Score	Top 25% Score	Top 10% Score
ATT (Borusyak, Jaravel, Spiess)	0.102***	1.820*	0.144***	0.080***
	(0.030)	(0.972)	(0.035)	(0.029)
Num. Obs	11357	11357	11357	11357
Control outcome mean	62.880	555.563	13.466	5.155
P-value pre-trends	0.229	0.794	0.294	0.752

Note: Standard errors (clustered at the program level) are in parentheses. The control outcome mean represents the average number of first-preference applications in the control group. All programs are offered by accredited institutions. The treatment group includes programs receiving accreditation for the first time. Average treatment effects on the treated follow Borusyak et al. (2024). * p<0.1, ** p<0.05, *** p<0.01.

Table 5: Estimated effects of accreditation on first-choice applications, by socioeconomic background

	Family income			Parental education		
	Low	Middle	High	No school	High school	College degree
ATT (Borusyak, Jaravel, Spiess)	0.085***	0.087***	0.105***	0.078**	0.114***	0.071**
	(0.032)	(0.032)	(0.033)	(0.032)	(0.031)	(0.031)
Num. Obs	11357	11357	11357	11357	11357	11357
Control outcome mean	25.406	24.943	14.532	8.735	27.551	24.573
P-value pre-trends	0.146	0.76	0.663	0.631	0.236	0.51

Note: Standard errors (clustered at the program level) are in parentheses. The control outcome mean is the average number of first-preference applications in the control group. All programs are offered by accredited institutions. The treatment group includes programs accredited for the first time. Average treatment effects on the treated follow Borusyak et al. (2024). * p<0.1, ** p<0.05, *** p<0.01.

Table 6: Student Enrollment and Persistence Responses to First-Time Accreditation

	Short-Term (1 yr)		Mid-Term (3 yrs)	
	Enrollment	Attrition	Transf out	Perm exits
ATT (Borusyak, Jaravel, Spiess)	0.073***	-0.010*	-0.010**	-0.003
	(0.026)	(0.005)	(0.004)	(0.004)
Num. Obs	11565	11532	11565	11565
Control outcome mean	61.279	0.232	0.122	0.069
P-value pre-trends	0.257	0.192	0.274	0.266

Note: Standard errors (clustered at the program level) are in parentheses. The control outcome mean represents the average number of students in the control group for each outcome. All programs are offered by accredited institutions. The treatment group includes programs accredited for the first time. Estimates for Lost Enrollment and Transfers exclude 2019–2020 data due to challenges in identifying dropouts and graduates for recent cohorts. Average treatment effects on the treated follow Borusyak et al. (2024). * p<0.1, ** p<0.05, *** p<0.01.

Table 7: Estimated Effects on Enrollment and Persistence by Institutional Quality

	Short-Term (1 yr)		Mid-Term (3 yrs)	
	Enrollment	Attrition	Transf out	Perm exits
Panel A: Implied Total Effects by Institutional Quality				
Baseline (β_1)	-	-0.012	-0.011**	0.014***
	-	(0.010)	(0.005)	(0.005)
Enchanced (β_1 + β_2)	-	-	-	-0.012***
	-	-	-	(0.001)
Top-tier (β_1 + β_3)	-	-	-0.003	-0.015***
	-	-	(0.005)	(0.002)
Num. Obs	10456	3628	5184	10456
Control Mean				
Baseline	-	0.263	0.075	0.096
Enhanced	-	-	-	0.055
Toptier	-	-	0.194	0.044
P-value Pre-trends				
Baseline	0.067	0.509	0.258	0.492
Enhanced	0.004	0.012	0.044	0.601
Top-tier	0.064	0.014	0.116	0.309

Note: First panel reports the implied effects of accreditation for each institutional category (Basic, Advanced, and Excellence). Standard errors are clustered at the program level. Control outcome means reflect the average in the control group by category. Accreditation levels follow CNA guidelines: Basic (reference), Advanced, and Excellence. Omitted coefficients are due to evidence of differential pre-trends for that institutional category, violating the parallel trends assumption, following the p-values for the F tests on pre-treands. Estimates follow Borusyak et al. (2024). * p<0.1, ** p<0.05, *** p<0.01.

Table 8: Estimated Effects on Enrollment and Persistence by Regulatory Framework

	Short-Term (1 yr)		Mid-Term (3 yrs)	
	Enrollment	Attrition	Transf out	Perm exits
Panel A: Implied Total Effects by Regulatory Framework				
Voluntary (β_1)	0.073***	-0.010*	-0.008**	-0.001
	(0.024)	(0.005)	(0.004)	(0.003)
Mandated (β_1 + β_2)	-0.201***	-0.012*	-0.009**	-0.003
	(0.017)	(0.004)	(0.002)	(0.003)
Num. Obs	12963	12935	12963	12963
Control Mean				
Voluntary	61.279	0.232	0.122	0.069
Mandatory	56.14	0.183	0.067	0.065
P-value Pre-trends				
Voluntary	0.257	0.192	0.274	0.266
Mandatory	0.306	0.183	0.492	0.323

Note: First panel reports the implied effects of accreditation for each regulatory framework (Voluntary or Mandatory). Standard errors are clustered at the program level. Control outcome means reflect the average in the control group by category. The p-values for the F tests on pre-trends serve as evidence of parallel trends assumption by category. Estimates follow Borusyak et al. (2024). * p<0.1, ** p<0.05, *** p<0.01.

Appendix

A The Chilean Accreditation System

This appendix details the institutional features of the Chilean accreditation system from 2007 – 2018, maps the official evaluation criteria to the variables used in our analysis, and discusses potential unobserved factors.

A.1 Evolution of the Chilean Accreditation System

The Chilean accreditation system has undergone significant changes since its inception as a pilot program. The table below summarizes key milestones in its development, including its official establishment, the role of private agencies during our study period, and recent reforms that have altered its structure. Understanding this evolution provides context for the specific regulatory environment (2007 – 2018) analyzed in this paper.

Table A.1: History of the higher education accreditation system in Chile

1990	National Education Council
1999 – 2000	National Commission of Undergraduate Accreditation (CNAP) Pilot Program One public agency – voluntary process
2006	Official System National Accreditation Commission (CNA) Private agencies are part of the system. CNA supervises
2018 – Present	CNA is now the only agency *Only institutions can be accredited, not programs.*[a]

[a] A recent Law approved in 2018 (link to the Law's description https://bcn.cl/2fcks) eliminated the accreditation of programs and privatization of the accreditation system, previously managed by private agencies. Additionally, the law now requires mandatory institutional accreditation. Furthermore, the law introduces stricter sanctions in cases of conflict of interest. It changes the process of appointing commissioners, under the authority of the President, who are responsible for overseeing the accreditation.

A.2 The Accreditation Process: From Application to Decision

The accreditation system formalized by the 2006 Quality Assurance Act is overseen by the National Commission of Accreditation (CNA). While the CNA oversees both institutional and program-level accreditation, our study focuses on the latter, which provides a relevant major-specific signal that students observe when making application decisions.

The program-level accreditation process follows a standardized, multi-stage procedure:

1. Self-Assessment: The process is initiated by the program itself. This first stage involves an extensive internal review where the program's faculty and administration compile a detailed self-assessment report. This report gathers quantitative and qualitative evidence to argue that the program meets the CNA's quality standards. The creation of this report is a lengthy process, and its timing is an endogenous choice made by the institution.

2. External Peer Review: After the self-assessment report is submitted, an external committee of peer evaluators (academics from other institutions) is assigned. This committee reviews the report and conducts a multi-day site visit to interview students, faculty, and administrators, and to inspect facilities.

3. Formal Decision: The peer review committee submits its own report to the CNA (or, during this period, a delegated private agency). The agency then makes a final judgment, weighing the program's self-assessment against the peer review report.

4. Issuance of Accreditation: The agency's final decision is a public announcement. Programs are granted an accreditation status for a specific number of years, ranging from zero (denial) to a maximum of seven.

The entire process, from initial request to final resolution, typically takes up to a year, possibly extending for a few more months. As shown in the main paper, these decisions are announced throughout the calendar year, with the majority occurring after the March application deadline, making them a clean information shock for the next application cycle. Reaccreditation requires repeating the entire process.

A.3 CNA Evaluation Criteria and Variable Mapping

A key part of our empirical strategy is to test whether universities invest in long-term structural inputs or focus on more malleable, outcome-based metrics. This choice is guided directly by the CNA's official evaluation dimensions. We use data from the Higher Education Information System (SIES) and the Ministry of Education to construct proxies for these dimensions.

The CNA framework assesses multiple areas, including "curriculum relevance, faculty qualifications, infrastructure, and student outcomes" (CNAP, 2007). The table below maps these criteria to the specific variables used in our institutional response analysis

Table A.2: CNA Evaluation Criteria and Variable Proxies

CNA Evaluation Dimension	**Paper's Proxy Variable(s)**	**Source**	**Construction**
1. Student Outcomes	On-Time Graduation Rate	Ministry of Education	Explicit CNA metric. Following CNA guidelines, it's defined as the share of a first-year cohort graduating in (nominal duration + 1) years.
2. Faculty Resources	Lecturers-per-Student Ratio	SIES	This proxies for "faculty sufficiency". We use annual faculty headcounts reported by the university, normalized by program enrollment.
3. Infrastructure	1. Buildings per Student 2. Campus Size per Student (sq. meters)	SIES	These proxies for "infrastructure" are normalized to reflect resource availability. Campus-level data is allocated to the program-level using an enrollment-weighted procedure and normalized by median program enrollment.
4. Other Program Metrics (Not explicit CNA criteria but highly visible)	1. Declared Capacity (Slots) 2. Annual Tuition	SIES	These are program-level metrics reported annually and made public prior to the admissions cycle.

Note: This table maps the official CNA evaluation dimensions to the proxy variables used in our institutional response analysis. All variable construction follows the descriptions in Section 3 and Appendix A. Variables for faculty and infrastructure are normalized per student to ensure comparability.

A.4 Institutional Features Relevant to Identification

Several specific features of the accreditation system during our study period are critical for our empirical strategy. First, when a university offers the same program across multiple campuses under a unified curriculum, accreditation is granted jointly for the program as a whole, applying uniformly to all locations. This institutional design is essential, as it ensures a single, uniform treatment date for all campus-level units within a given program, simplifying the staggered adoption setup. Second, the empirical relationship between the duration of program-level accreditation and institutional accreditation is relatively weak (correlation of 0.18), allowing for substantial variation in program-level status within institutions. This enables our difference-in-differences approach, using program fixed effects, to isolate the impact of program-specific signals while implicitly holding broader institutional quality constant. Finally, the distinction between voluntary and mandatory accreditation regimes provides valuable variation. While our main analysis focuses on the voluntary setting where self-selection is informative, we leverage the mandatory regime in our heterogeneity analysis (Section 6) to examine the signal's effect when this selection mechanism is absent. Figures A.1 and 1 illustrate the staggered adoption patterns under both regimes.

A.5 Signal Visibility and Public Disclosure

For accreditation to influence student choice, the final signal must be observable. The Chilean system ensures this through mandated public disclosure of the accreditation outcome. The CNA requires that the final accreditation decision, including the duration in years granted (from 0 to 7), be made public (CNAP, 2007). Institutions then actively use this official status in their marketing efforts, disseminating it through CNA channels, institutional websites, and various promotional materials like brochures, billboards, and university fairs (see Appendix Figures A.3, A.5, and A.4 for examples).

This structure ensures that the final accreditation status functions as an observable quality signal for students making decisions during the December-January application cycle. While institutions might voluntarily signal that they are undergoing evaluation before a decision is reached, the crucial information, such as the specific outcome and duration, remains uncertain until the official public announcement. This residual uncertainty is key to our identification strategy: for prospective students, the release of a new accreditation status acts as a largely unanticipated information shock, and for currently enrolled students, the outcome also represents new information about the validated quality of their programs.

A.6 Unobserved Factors and Limitations

Our empirical analysis relies on the observable proxies detailed in Table A.2. However, the CNA's evaluation framework is holistic and includes qualitative dimensions and institutional processes that our data cannot capture. This leads to several limitations regarding the measurement of institutional response.

1. Unobserved Quality of Inputs: Our measures for infrastructure and faculty (e.g., buildings per student, lecturers-per-student ratio) are quantitative. The CNA's assessment of "infrastructure" or "faculty sufficiency" is also qualitative. Our data do not capture unobserved, time-varying changes in the quality of these inputs, such as building renovations, upgrades to IT infrastructure, improvements to library databases, or the hiring of more skilled (but not more numerous) faculty. If programs invest in these unobserved quality margins, our analysis would understate the extent of their "long-term, structural investment," potentially overstating the relative

importance of the malleable metrics we observe.

2. Unobserved Investments and Expenditures: We do not observe detailed, program-level expenditures. Universities could be making other "long-term, structural investments" in areas not proxied by our variables, such as curriculum redesign, new student support services (like career counseling), or investments in technology. As with unobserved quality, this data limitation means our estimates of "long-term" investment are a lower bound.

3. Endogeneity of Application Timing: As stated in the main paper, a program's decision to seek accreditation is endogenous. While our identification strategy relies on the announcement of the outcome being an unanticipated information shock to students, the timing of the application is a strategic choice made by the program. This choice can be driven by unobserved, time-varying factors such as the arrival of a new program director, a response to a competitor's accreditation, or internal performance trends. If these unobserved factors are also correlated with trends in our outcome variables (e.g., a new director both applies for accreditation and implements policies that improve graduation rates), it would confound our interpretation of the pre-treatment coefficients. Our event-study design, which explicitly tests for pre-trends, is intended to detect and control for this, but the endogeneity of the timing remains a challenge.

B Hypotheses and Related Literature

Markets for higher education are intrinsically shaped by informational frictions. Prospective students, when choosing a college or major, make high-stakes investment decisions under deep uncertainty about program quality, institutional fit, and future labor market returns (Dillon and Smith, 2020; Lovenheim and Smith, 2022; Stange, 2012). This problem of asymmetric information, first formalized in the "market for lemons" by Akerlof (1970), is particularly relevant in systems with heterogeneous providers and weak prior signals of quality (Hoxby, 2002; MacLeod and Urquiola, 2015). In such settings, formal third-party certification, such as program-level accreditation, can serve as a valuable mechanism for mitigating these information asymmetries and improving market efficiency (Dranove and Jin, 2010). This paper investigates the extent to which accreditation fulfills this role by analyzing how students in Chile's university education system respond to the provision of this new quality signal.

This section develops the key hypotheses that guide our empirical analysis, grounding each in the relevant economic literature. We first hypothesize that university programs, acting as strategic agents, will engage in "window dressing" by manipulating malleable, low-cost metrics rather than making deep structural investments. This hypothesis stems from the literature on quality disclosure, which highlights the incentives for firms to selectively present information. Second, we test the hypothesis that students will respond positively to the accreditation signal, increasing their demand for newly certified programs. This prediction is rooted in the literature on markets with information frictions, where even a potentially gamed signal can reduce uncertainty for consumers. Finally, we develop hypotheses regarding the heterogeneity of these effects. Drawing from theories on the interaction between formal and informal signals, we test whether accreditation complements or substitutes for existing institutional reputation. Informed by studies on equity and information interventions, we also hypothesize that the simplicity of the accreditation signal will make it broadly accessible across socioeconomic groups.

B.1 Higher Education as an Investment under Uncertainty and Learning

We begin by conceptualizing higher education as an experience good, where its true quality and match-specific value are revealed to students only after enrollment and consumption (Nelson, 1970). This temporal resolution of uncertainty is a central feature of the student experience. From the student's perspective, application and enrollment decisions must be made based on noisy priors formed from observable but imperfect signals like institutional branding, tuition levels, and informal peer networks (Buss et al., 2004; Perna, 2006). Critical institutional features, such as instructional quality or labor market connections, are difficult to observe ex-ante and are not always credibly signaled by providers.

A rich body of empirical work confirms that students learn about program quality and their own academic fit over time and adjust their behavior accordingly (Aina et al., 2022; Kirp, 2019). This literature underscores that learning after enrollment can be costly, generating inefficiencies that credible ex-ante information could prevent. For instance, Stinebrickner and Stinebrickner (2012) provides evidence that college dropout is often a rational response to new information; students who entered with overly optimistic beliefs about their academic ability update these beliefs after receiving their first-year grades and subsequently revise their enrollment decisions. Recently, Arcidiacono et al. (2025) estimate a dynamic model of college attendance and find that information frictions regarding academic ability and labor market productivity are major drivers of attrition. They show that removing these frictions would increase four-year college graduation rates by 4.4 percentage points and significantly improve ability sorting in the labor market. The sequential nature of these decisions creates an "option value" of college attendance, as students can preserve the option to continue their education after learning more about their tastes and aptitude (Arcidiacono, 2004; Arcidiacono et al., 2025; Bordon and Fu, 2015; Stange, 2012).

The common thread in these models is that post-enrollment learning is an integral, yet potentially costly, part of the higher education process. This literature provides the foundation for our first key hypothesis. We hypothesize that the introduction of a formal quality signal, such as accreditation, will significantly alter student application and enrollment behavior by reducing ex-ante uncertainty. Specifically, we predict that by providing credible new information, accreditation will cause an increase in student demand for newly certified programs. Furthermore, by allowing for better-informed initial choices, we expect the signal to improve the quality of the student-program match, leading to a reduction in costly post-enrollment adjustments like program switching and dropout (Dillon and Smith, 2020).

B.2 Accreditation as a Mechanism for Information Disclosure

Given this environment of uncertainty, accreditation can be interpreted through the primary economic theories of information transmission: costly signaling and information disclosure.

First, voluntary accreditation fits naturally into the signaling framework of Spence (1973). We can conceptualize accreditation as a costly signal that higher-quality programs are more likely to pursue, given the substantial direct and indirect costs associated with the compliance and evaluation process, as well as the expected reputational payoff. In this framework, programs endogenously decide whether to undergo accreditation, ideally creating a separating equilibrium in which only sufficiently high-quality programs self-select into certification (Dranove and Jin, 2010). This logic applies most directly to the voluntary accreditation regime that we study in Chile.

Second, accreditation can be viewed as a disclosure mechanism. Classical "unraveling" models predict that, in the absence of frictions, high-quality producers will voluntarily disclose their type to separate from lower-quality competitors, leading to full disclosure (Grossman, 1981; Milgrom, 1981). However, as previous reseach shows, full unraveling rarely happens in practice (Cameron et al., 2023; Dranove et al., 2003; Jin, 2005; Vatter, 2023). Frictions, such as heterogeneous consumer sophistication, costs associated with verifying and disclosing information, and strategic incentives to withhold or conceal, often lead to partial disclosure equilibria (Jin et al., 2021; Leuz et al., 2008). This failure of voluntary disclosure provides a strong rationale for mandatory certification systems. A powerful empirical demonstration of the impact of disclosure comes from Jin and Leslie (2003), who studied the introduction of mandatory hygiene grade cards for restaurants. They find that making quality information credible and easily comparable not only caused consumer demand to become sensitive to hygiene scores, but also induced restaurants to make substantive improvements to hygiene quality, leading to a 20% reduction in hospitalizations for foodborne illness. This illustrates that a well-designed disclosure system can profoundly alter both consumer and firm behavior, generating significant welfare gains.

While these theories suggest accreditation should ideally separate high-quality programs, they also highlight the strategic incentives firms face when the signal is being evaluated. The high cost of genuine structural improvements, combined with the significant reputational payoff from a positive signal, can lead programs to pursue less costly strategies to influence the outcome. This leads to our second key hypothesis. We presume programs engage in strategic "window dressing" prior to their evaluation. Specifically, we predict that rather than undertaking deep, long-term investments in faculty or infrastructure, programs will focus on improving more malleable and visible metrics that are explicit inputs in the accreditation assessment, such as on-time graduation rates. This behavior represents an equilibrium outcome where the signal's content is endogenously shaped by the institutions themselves.

B.3 The Interaction of Signals: Formal Certification and Endogenous Reputation

Accreditation is only one of several signals available in the higher education market. Institutional reputation, developed over time through sustained performance, research output, and alumni networks, provides a powerful alternative signal of quality. However, it may be too general to infer a particular program's quality (Hörner, 2002; MacLeod and Urquiola, 2015). This raises a critical question for our analysis: how do formal certification and informal, long-run reputation interact? Are they substitutes or complements?

Theory suggests the relationship depends on whether the signals provide redundant or distinct information. Signals are substitutes when they convey similar information about the same underlying quality dimension, making one less valuable when the other is present (Elfenbein et al., 2012, 2015). For example, a formal certification may be most useful for firms with weak informal reputations (Bar-Isaac and Tadelis, 2008; Dranove and Jin, 2010; MacLeod and Urquiola, 2015). Compelling empirical evidence is presented in Elfenbein et al. (2012), who study charitable giving on eBay. They find that the "charity premium" is largest for new sellers with no established feedback record and diminishes as a seller's reputation grows. This suggests that alternative signals are most valuable when reputational signs are weak or absent.

However, signals can be complements if they provide information on different quality dimensions or if one signal helps to interpret the other (Börgers et al., 2013). This may be particularly relevant in complex markets like higher education, where a general signal of institutional reputation may be too coarse for students making specific program choices. A targeted signal, like program-level accreditation, could help students better interpret or trust the broader institutional signal. Empirically, Colombo et al. (2019) find evidence of such complementarity in the market for science-based

IPOs, where signals of scientific quality (affiliation to a prestigious university) and financial quality (prestigious underwriter backing) are additive, as they reduce uncertainty in different domains.

This theoretical tension leads to our third set of competing hypotheses. If accreditation serves as a substitute for pre-existing signals, its impact will be greatest for programs at less reputable institutions. On the other hand, if their relationship is of complements, then the effect of accreditation will be largest for programs at more reputable institutions.

Finally, the signal's value is also shaped by the regulatory framework. The signaling and disclosure theories discussed previously both rely on a firm's choice to reveal information. In the voluntary regime, the value of the signal comes from both the accreditation outcome and the act of self-selection into the process (Jin and Leslie, 2003; Spence, 1973). In the mandatory regime, this self-selection mechanism is eliminated, potentially reducing the signal's informational content (Jin, 2005). This leads to our fourth hypothesis. The effect of accreditation on student outcomes will be significantly weaker under a mandatory regime than under a voluntary one.

B.4 Student Responsiveness to Information and Equity Implications

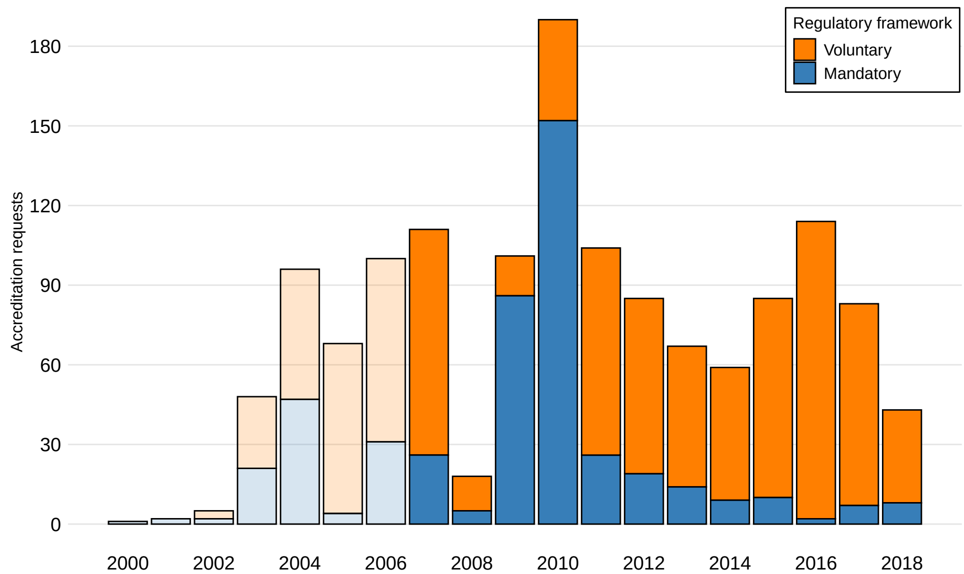
Ultimately, the empirical relevance of accreditation depends on whether students observe the signal and respond to it. A growing body of research, largely based on information-provision experiments, shows that students are indeed sensitive to new information about institutional quality and returns (Andrabi et al., 2017; Bettinger et al., 2012; Jensen, 2010). For instance, Wiswall and Zafar (2015) use experimental variation to show that providing students with data on earnings and program characteristics alters their beliefs and intended major choices. Similarly, Hastings et al. (2015) find that disclosing earnings information reduces demand for low-return degree programs in Chile.

While this body of work demonstrates the potential for information to change behavior, it largely relies on researcher-provided information in controlled settings. Our study contributes by evaluating the effects of an existing, institutionalized certification system operating at market scale. Of particular relevance is the work by Hoxby and Turner (2015), who demonstrate that providing customized, low-cost college guidance to high-achieving, low-income students substantially increases their likelihood of applying to and enrolling in high-quality, selective institutions. Their work powerfully illustrates that information frictions are not uniformly distributed, as they are particularly severe for students from disadvantaged backgrounds who may lack access to sophisticated guidance networks (Bergman, 2021; Hastings and Weinstein, 2008).

This raises a critical equity question. Can a public signal like accreditation overcome these barriers, or does the ability to interpret and act on the signal correlate with socioeconomic status? One perspective, drawing from life-cycle models of skill formation, suggests that the ability to process new information may itself be a skill correlated with socioeconomic background, potentially leading to larger effects for more advantaged students (Heckman and Mosso, 2014). However, an alternative perspective argues that simple, credible, and public signals can be particularly powerful for disadvantaged groups. Research by Dynarski et al. (2021) and Vatter (2023) suggests that simplifying complex information can disproportionately benefit those who lack access to private guidance or the resources to navigate complex information environments. This leads to our fifth hypothesis. The accreditation signal will be broadly accessible, and its effects on student applications and enrollment will be positive and significant across all socioeconomic groups, including for students from low-income backgrounds.

Evolution of First Accreditation Requests

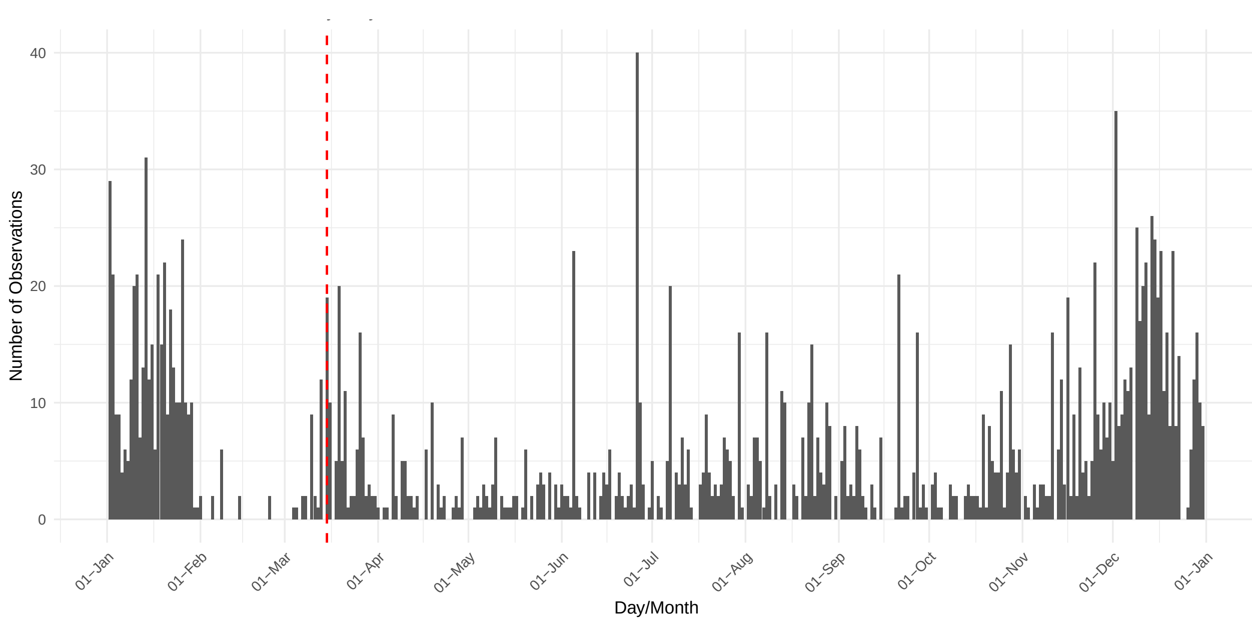
Figure A.1: Accreditation Requests by Regime



Note: This figure illustrates the evolution of first-time accreditation requests from 2000 to 2018, disaggregated by regulatory framework. The initial years (light orange bars, pre-2006) represent a pilot phase of the voluntary system. The officialization of the system in 2006 led to a significant increase in requests.

Distribution of notification dates

Figure A.2: Distribution of Date of Notification to Programs



Note: This figure plots the distribution of accreditation notification dates throughout the calendar year. The vertical line marks the start of the academic year in early March. Approximately 85% of notifications are issued after this date, confirming that the new accreditation status primarily influences the subsequent application cycle.

Institutions and programs advertising accreditation status

Figure A.3: Advertisement – Student College Fair

Note: A booth for the Universidad Academia de Humanismo Cristiano at a higher education fair. The indication, highlighted in green, advertises the 4-year accreditation status to prospective students.

Figure A.4: Advertisement – Subway Panel

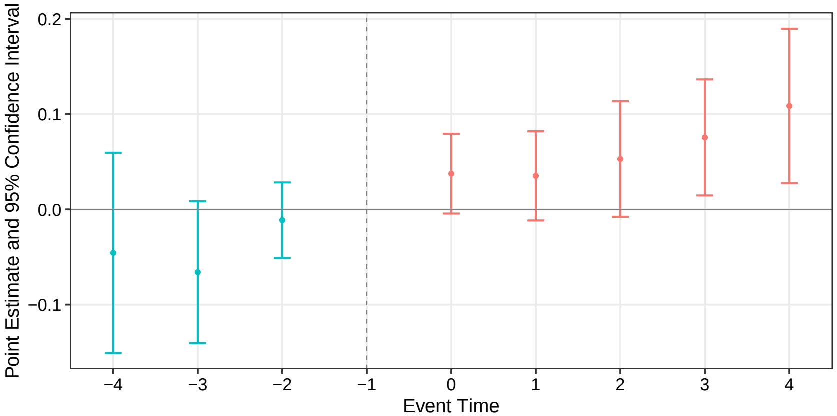
Note: An advertisement for the professional institute Duoc UC in a Santiago, Chile subway station. The highlighted area emphasizes the 7-year "Level of Excellence" accreditation, the highest possible duration.

Figure A.5: Advertisement – Street Sign

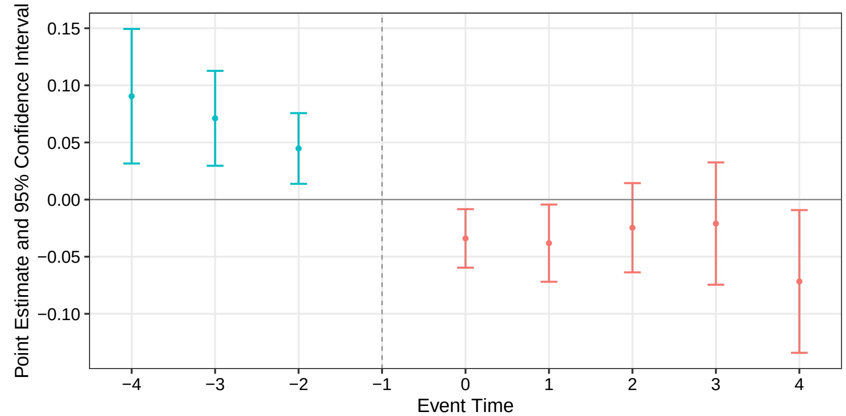
Note: A 2008 admissions billboard for the professional institute La Araucana. The highlighted portion prominently displays the official “Acreditado” (Accredited) seal from the National Accreditation Commission (CNAP).

Event study plots: Programs response by institutional quality

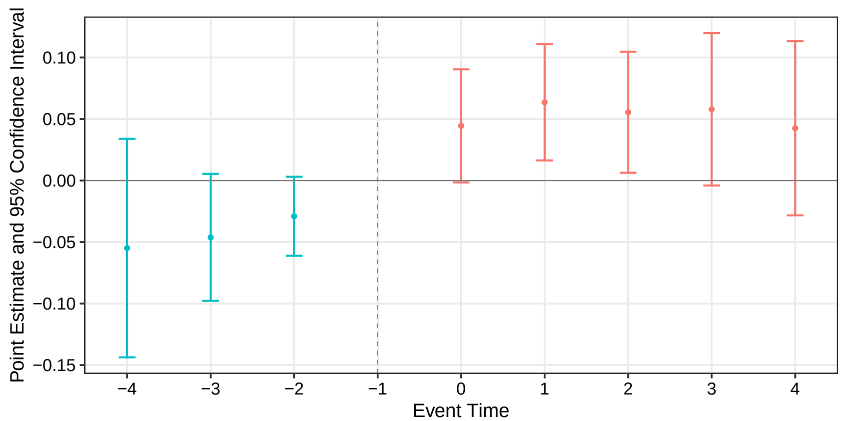
Figure A.6: Event study estimates: declared slots by institutional quality.



(a) Baseline quality



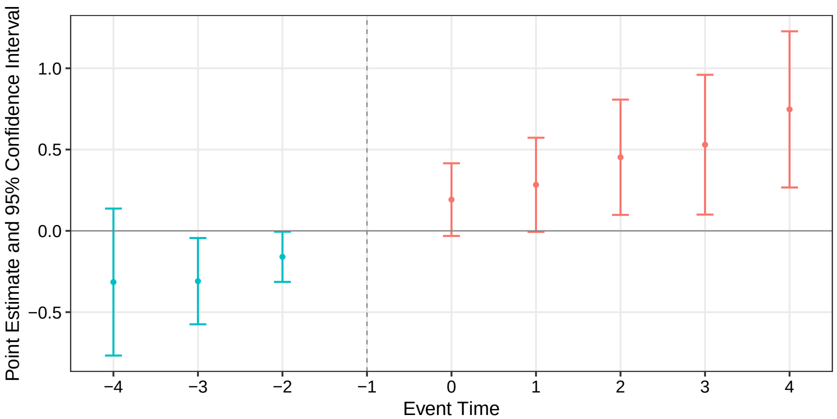
(b) Enhanced quality



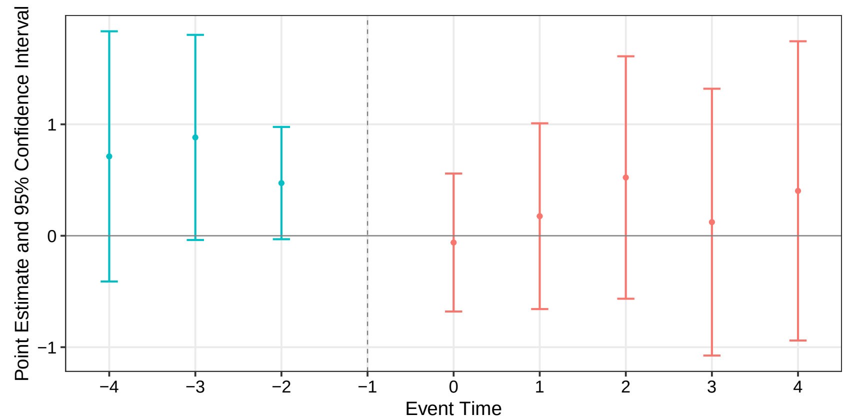
(c) Top-tier quality

Note: This figure plots event study estimates for the effect of first-time accreditation on declared slots, disaggregated by three tiers of institutional quality following the CNA guidelines: (a) Baseline quality, (b) Enhanced quality, and (c) Top-tier quality. The year prior to treatment ($t-1$) is the omitted reference category. Vertical bars represent 95% confidence intervals. Estimates follow Borusyak et al. (2024) with standard errors clustered at the program level.

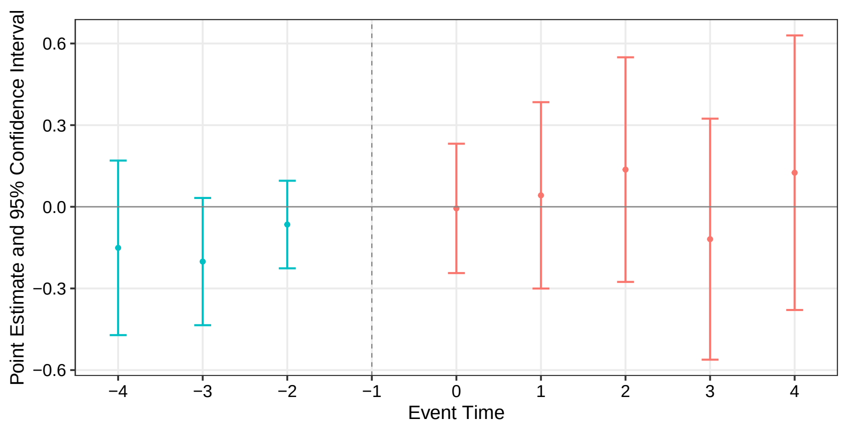
Figure A.7: Event study estimates: number of buildings by institutional quality.



(a) Baseline quality



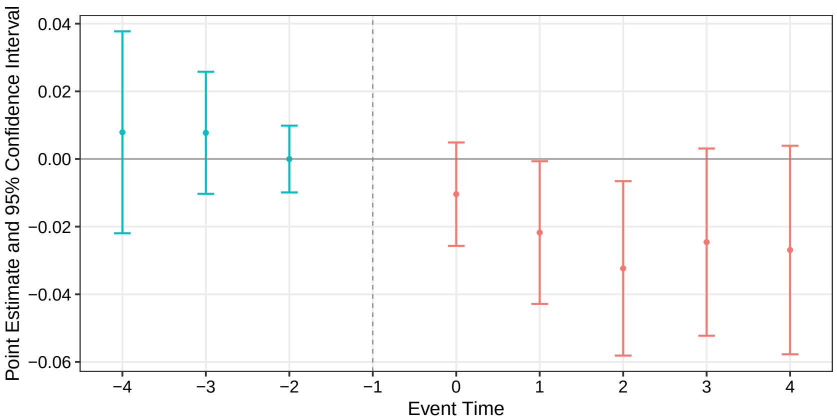
(b) Enhanced quality



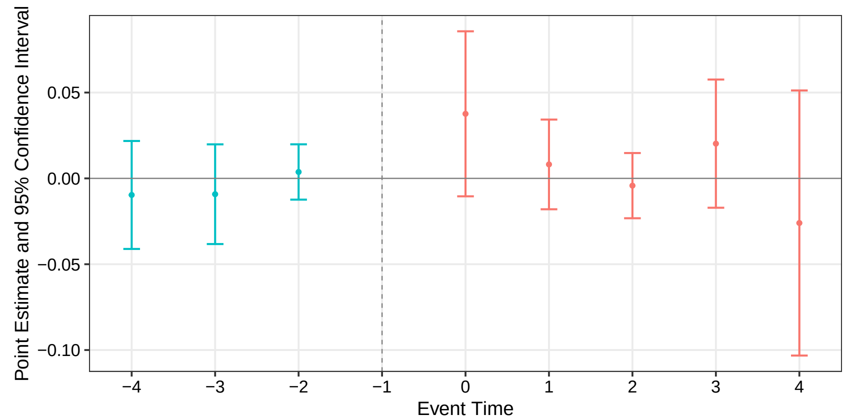
(c) Top-tier quality

Note: This figure plots event study estimates for the effect of first-time accreditation on buildings in campus, disaggregated by three tiers of institutional quality following the CNA guidelines: (a) Baseline quality, (b) Enhanced quality, and (c) Top-tier quality. The year prior to treatment ($t-1$) is the omitted reference category. Vertical bars represent 95% confidence intervals. Estimates follow Borusyak et al. (2024) with standard errors clustered at the program level.

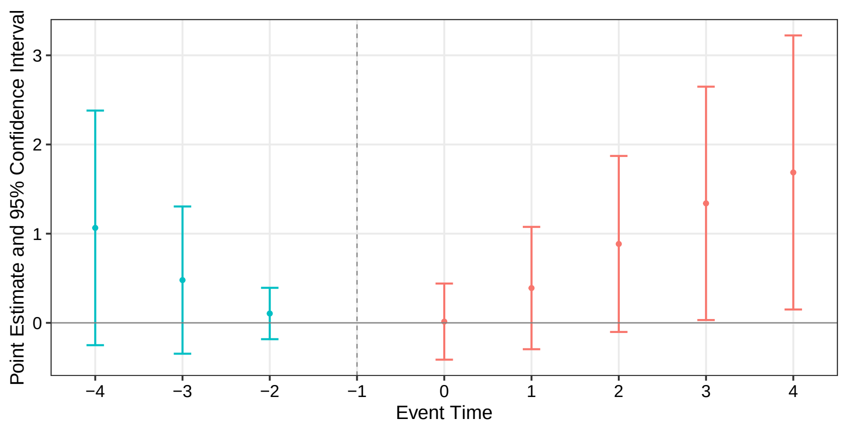
Figure A.8: Event study estimates: campus size by institutional quality.



(a) Baseline quality



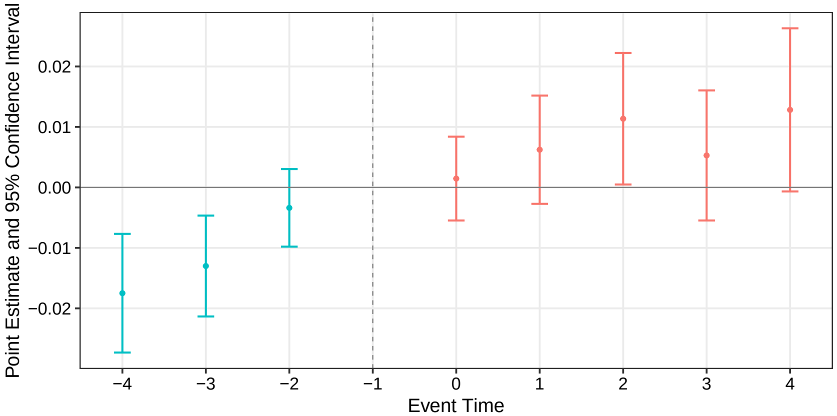
(b) Enhanced quality



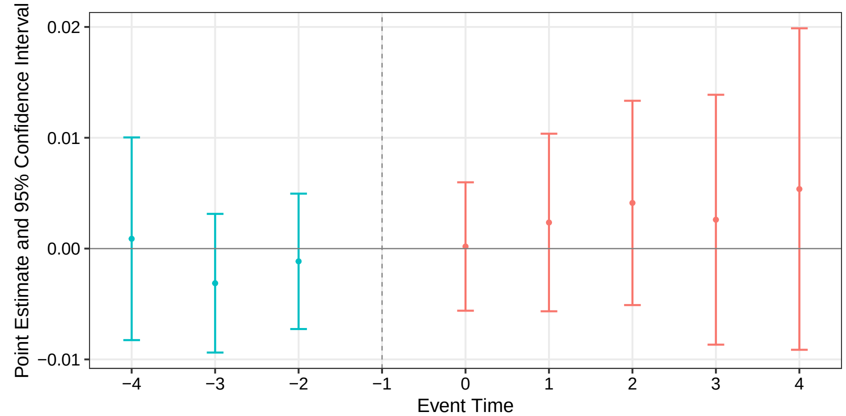
(c) Top-tier quality

Note: This figure plots event study estimates for the effect of first-time accreditation on campus size, disaggregated by three tiers of institutional quality following the CNA guidelines: (a) Baseline quality, (b) Enhanced quality, and (c) Top-tier quality. The year prior to treatment ($t-1$) is the omitted reference category. Vertical bars represent 95% confidence intervals. Estimates follow Borusyak et al. (2024) with standard errors clustered at the program level.

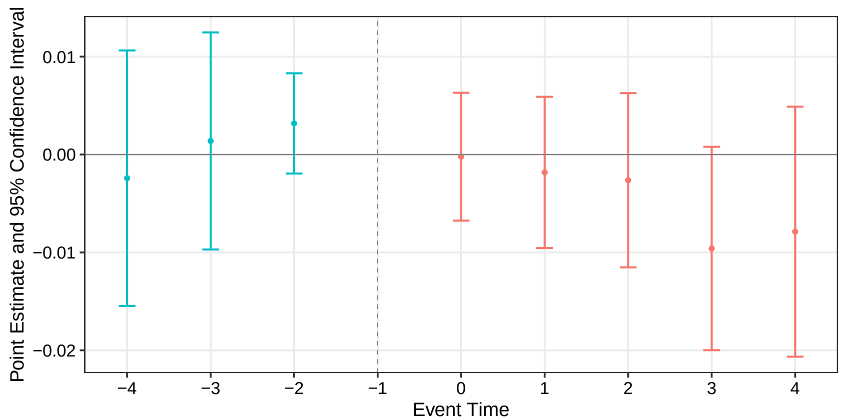
Figure A.9: Event study estimates: declared lecturers by institutional quality.



(a) Baseline quality



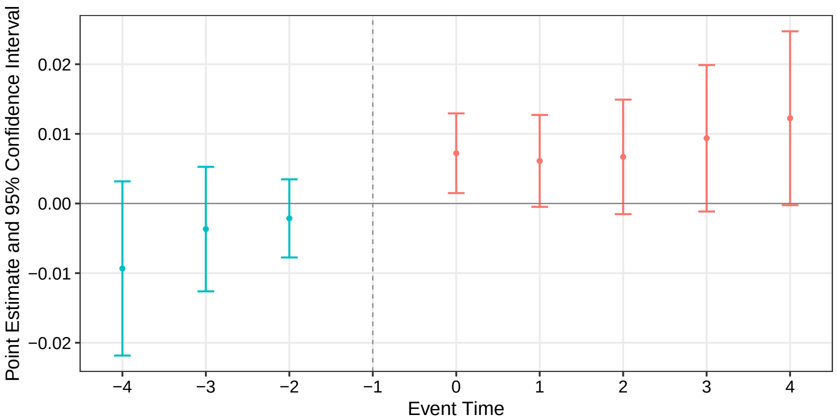
(b) Enhanced quality



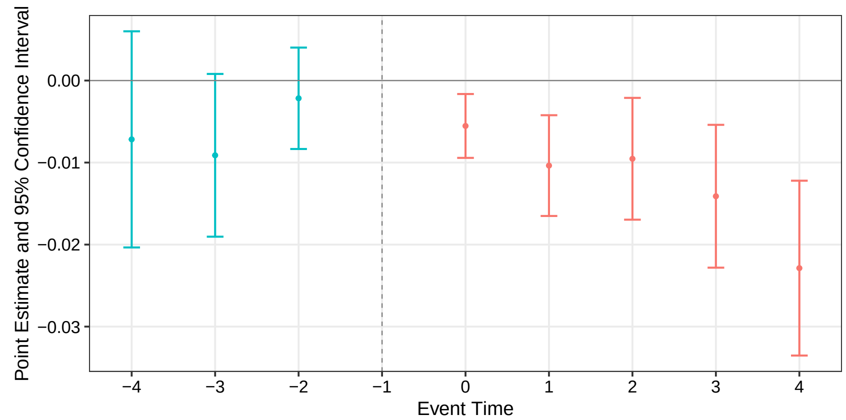
(c) Top-tier quality

Note: This figure plots event study estimates for the effect of first-time accreditation on declared lecturers, disaggregated by three tiers of institutional quality following the CNA guidelines: (a) Baseline quality, (b) Enhanced quality, and (c) Top-tier quality. The year prior to treatment ($t - 1$) is the omitted reference category. Vertical bars represent 95% confidence intervals. Estimates follow Borusyak et al. (2024) with standard errors clustered at the program level.

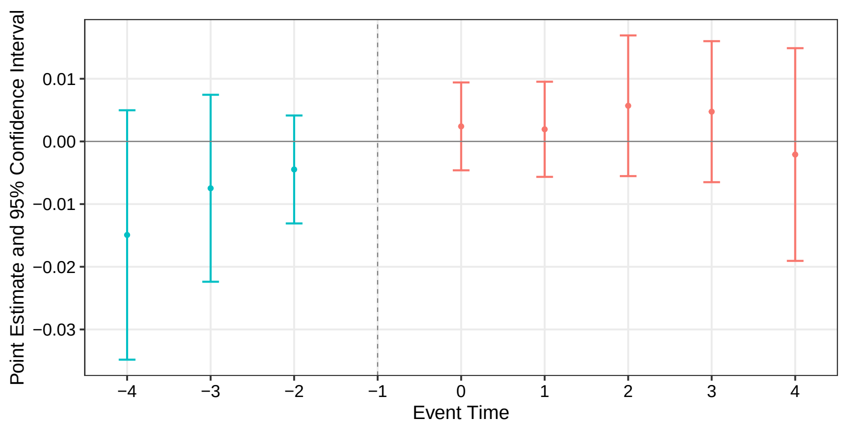
Figure A.10: Event study estimates: tuition by institutional quality.



(a) Baseline quality



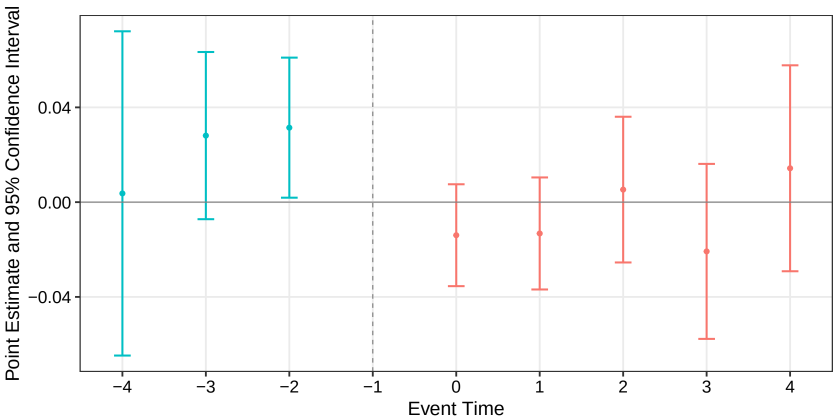
(b) Enhanced quality



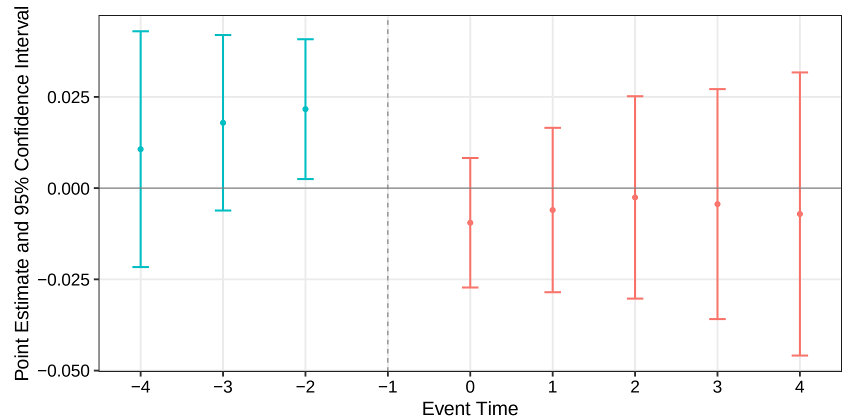
(c) Top-tier quality

Note: This figure plots event study estimates for the effect of first-time accreditation on tuition, disaggregated by three tiers of institutional quality following the CNA guidelines: (a) Baseline quality, (b) Enhanced quality, and (c) Top-tier quality. The year prior to treatment $(t-1)$ is the omitted reference category. Vertical bars represent 95% confidence intervals. Estimates follow Borusyak et al. (2024) with standard errors clustered at the program level.

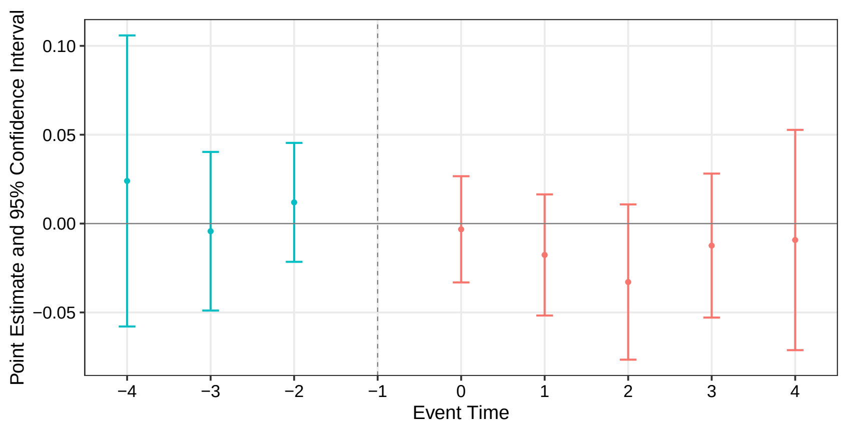
Figure A.11: Event study estimates: on-time graduates by institutional quality.



(a) Baseline quality



(b) Enhanced quality

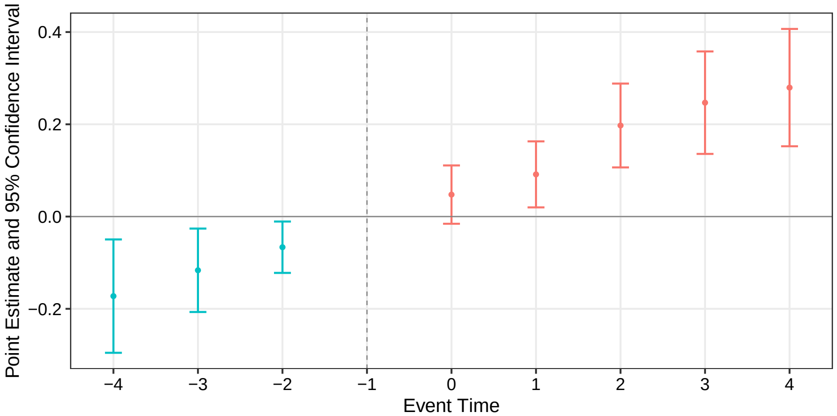


(c) Top-tier quality

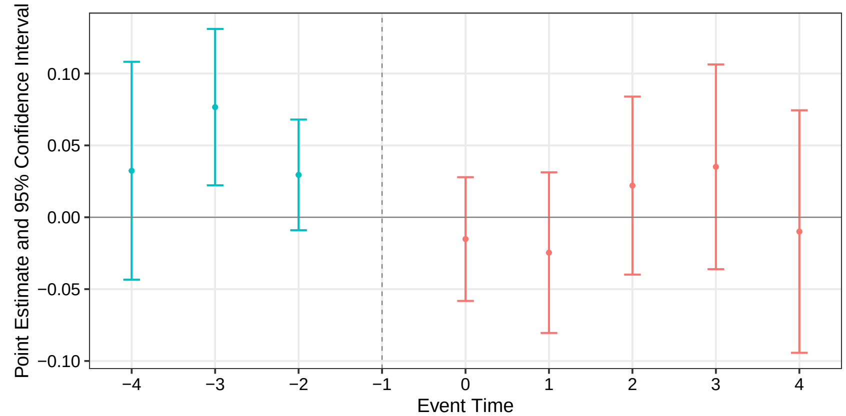
Note: This figure plots event study estimates for the effect of first-time accreditation on on-time graduation, disaggregated by three tiers of institutional quality following the CNA guidelines: (a) Baseline quality, (b) Enhanced quality, and (c) Top-tier quality. The year prior to treatment ($t-1$) is the omitted reference category. Vertical bars represent 95% confidence intervals. Estimates follow Borusyak et al. (2024) with standard errors clustered at the program level.

Event study plots: Students enrollment and persistence by institutional quality

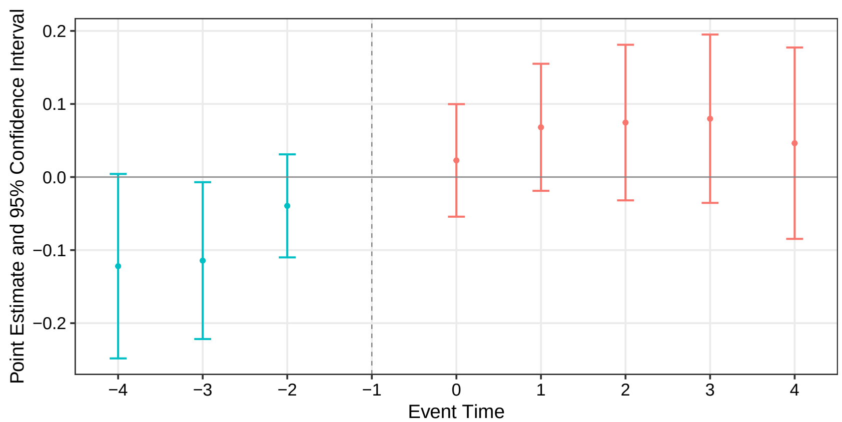
Figure A.12: Event study estimates: first-year enrollment by institutional quality.



(a) Baseline quality



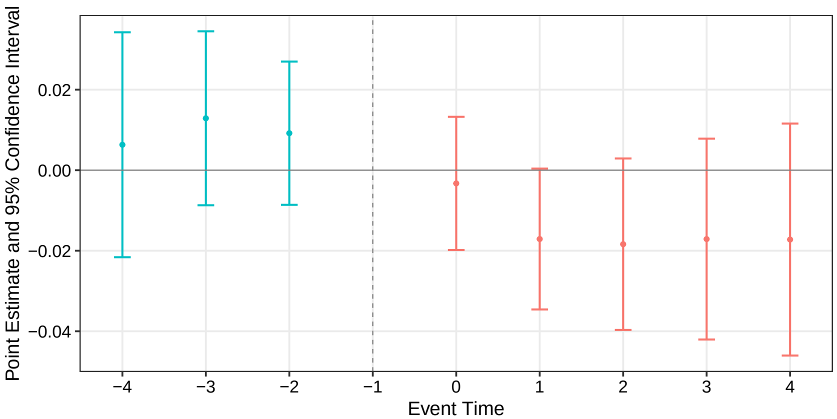
(b) Enhanced quality



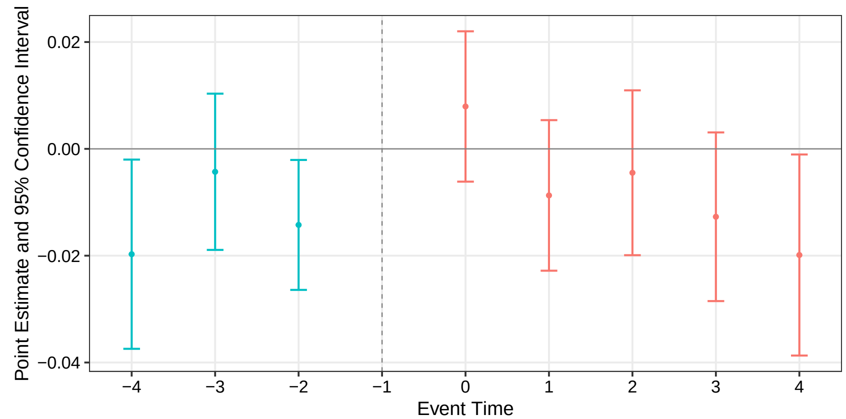
(c) Top-tier quality

Note: This figure plots event study estimates for the effect of first-time accreditation on first-year enrollment, disaggregated by three tiers of institutional quality following the CNA guidelines: (a) Baseline quality, (b) Enhanced quality, and (c) Top-tier quality. The year prior to treatment ($t-1$) is the omitted reference category. Vertical bars represent 95% confidence intervals. Estimates follow Borusyak et al. (2024) with standard errors clustered at the program level.

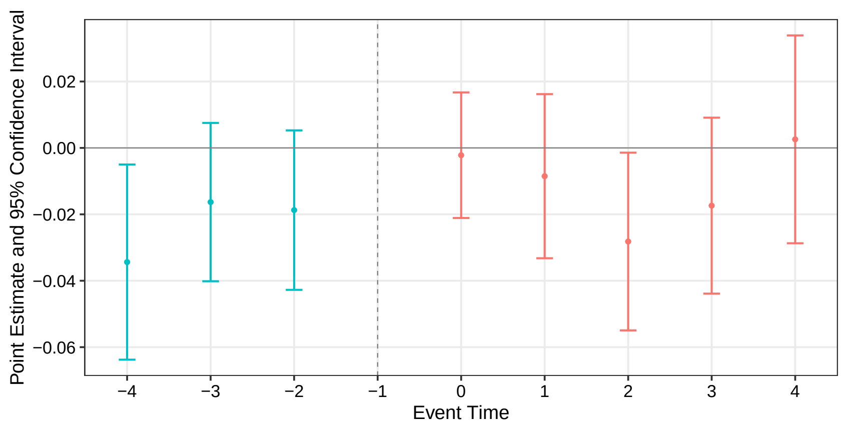
Figure A.13: Event study estimates: first-year attrition by institutional quality.



(a) Baseline quality



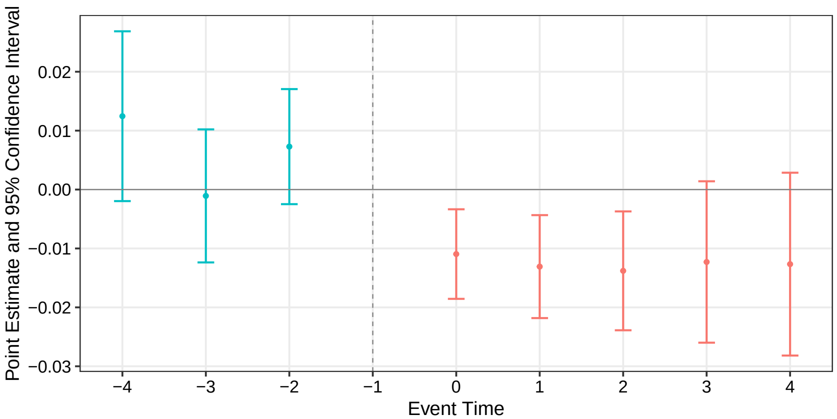
(b) Enhanced quality



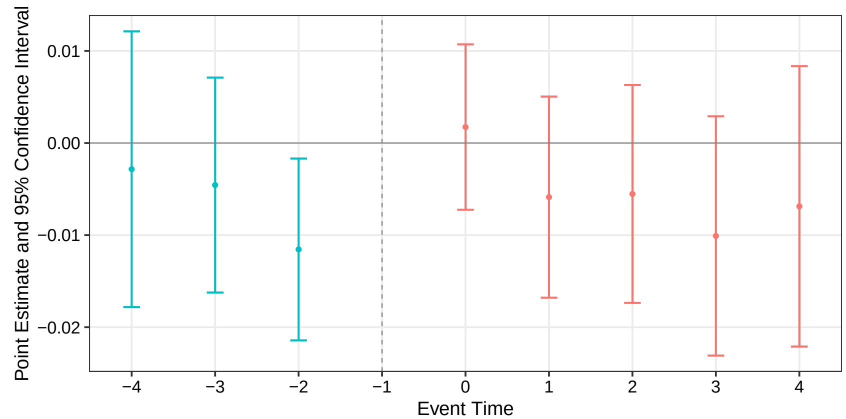
(c) Top-tier quality

Note: This figure plots event study estimates for the effect of first-time accreditation on first-year attrition, disaggregated by three tiers of institutional quality following the CNA guidelines: (a) Baseline quality, (b) Enhanced quality, and (c) Top-tier quality. The year prior to treatment ($t-1$) is the omitted reference category. Vertical bars represent 95% confidence intervals. Estimates follow Borusyak et al. (2024) with standard errors clustered at the program level.

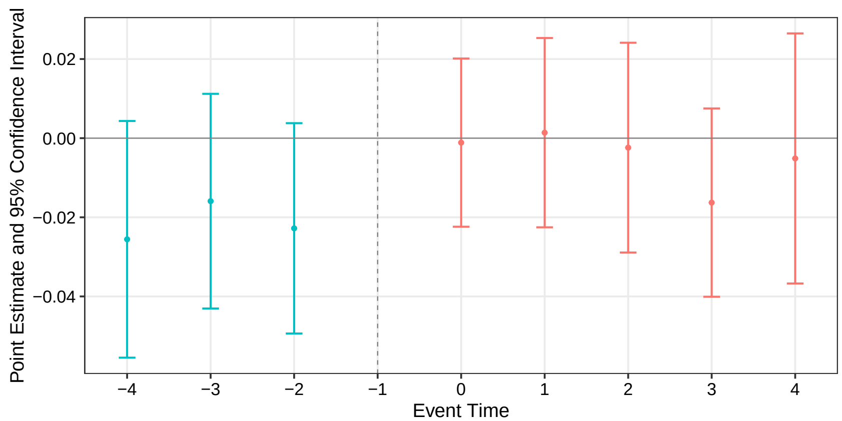
Figure A.14: Event study estimates: transfers-out by institutional quality.



(a) Baseline quality



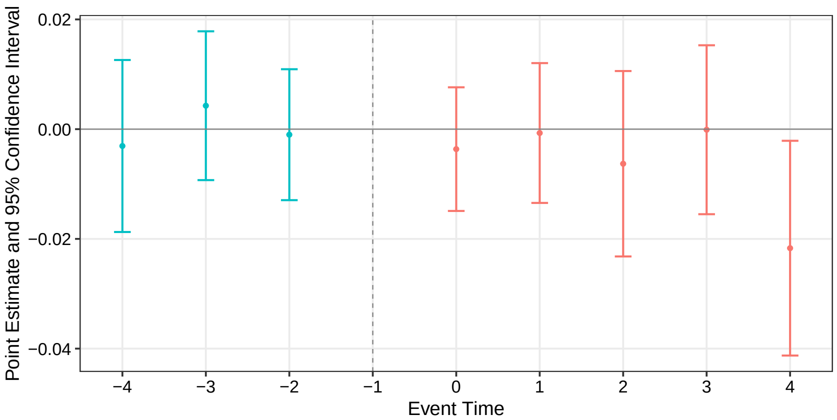
(b) Enhanced quality



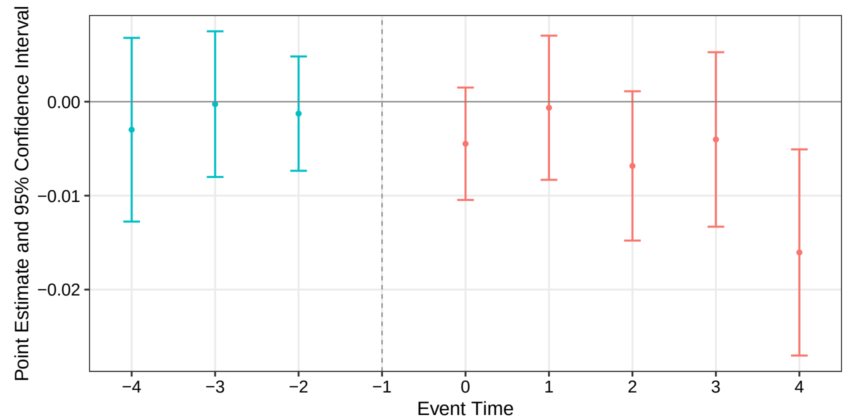
(c) Top-tier quality

Note: This figure plots event study estimates for the effect of first-time accreditation on trasfers to other institutions, disaggregated by three tiers of institutional quality following the CNA guidelines: (a) Baseline quality, (b) Enhanced quality, and (c) Top-tier quality. The year prior to treatment $(t-1)$ is the omitted reference category. Vertical bars represent 95% confidence intervals. Estimates follow Borusyak et al. (2024) with standard errors clustered at the program level.

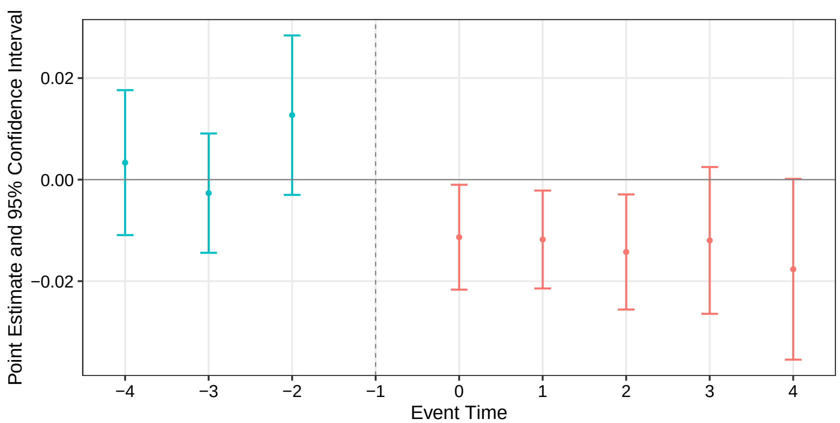
Figure A.15: Event study estimates: permanent exits by institutional quality.



(a) Baseline quality



(b) Enhanced quality

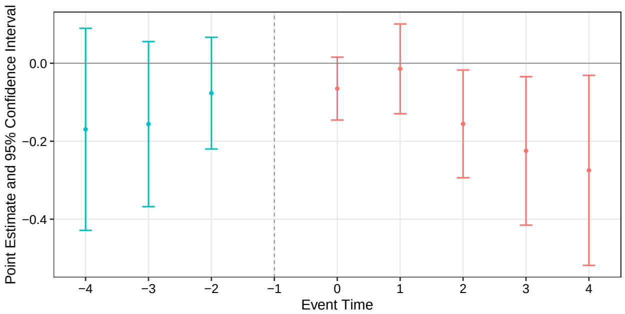


(c) Top-tier quality

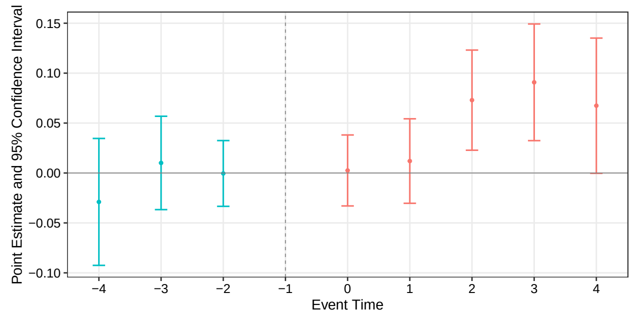
Note: This figure plots event study estimates for the effect of first-time accreditation on permanent exits, disaggregated by three tiers of institutional quality following the CNA guidelines: (a) Baseline quality, (b) Enhanced quality, and (c) Top-tier quality. The year prior to treatment ($t-1$) is the omitted reference category. Vertical bars represent 95% confidence intervals. Estimates follow Borusyak et al. (2024) with standard errors clustered at the program level.

Event study plots: Students enrollment and persistence by regulatory framework

Figure A.16: Event study estimates: first-year enrollment by regulatory framework.



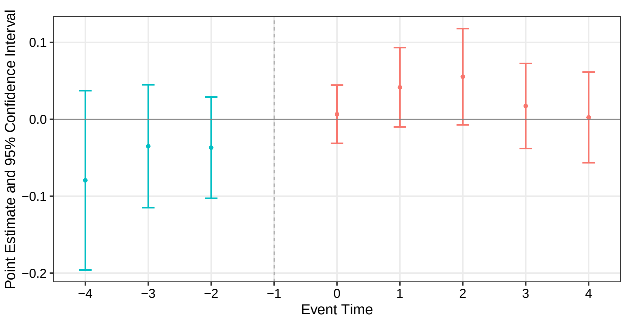
(a) Mandatory framework



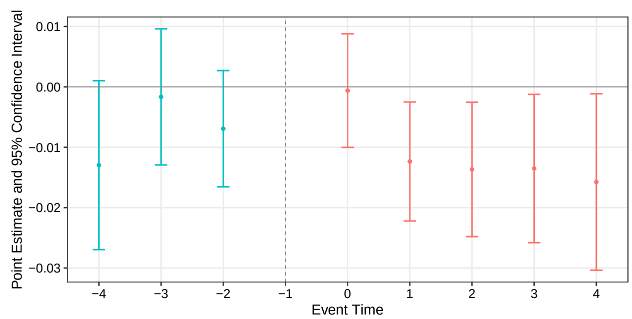
(b) Voluntary framework

Note: This figure plots event study estimates for the effect of first-time accreditation on first-year enrollment, disaggregated by regulatory framework: (a) mandatory framework and (b) voluntary framework. The year prior to treatment ($t-1$) is the omitted reference category. Vertical bars represent 95% confidence intervals. Estimates follow Borusyak et al. (2024) with standard errors clustered at the program level.

Figure A.17: Event study estimates: first-year attrition by regulatory framework.



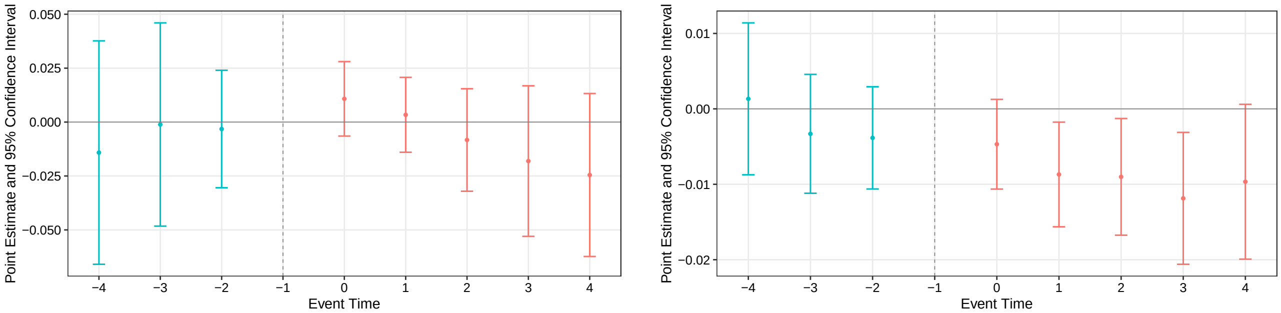
(a) Mandatory framework



(b) Voluntary framework

Note: This figure plots event study estimates for the effect of first-time accreditation on first-year attrition, disaggregated by regulatory framework: (a) mandatory framework and (b) voluntary framework. The year prior to treatment ($t-1$) is the omitted reference category. Vertical bars represent 95% confidence intervals. Estimates follow Borusyak et al. (2024) with standard errors clustered at the program level.

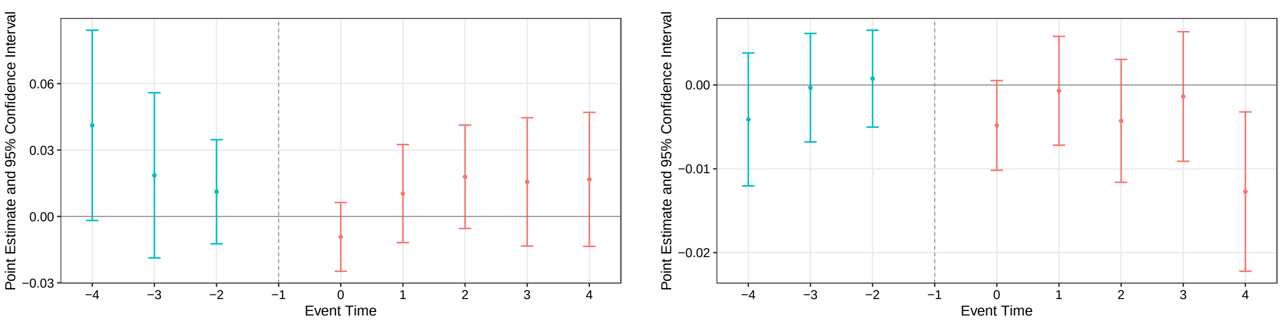
Figure A.18: Event study estimates: transfers-out by regulatory framework.



(a) Mandatory framework (b) Voluntary framework

Note: This figure plots event study estimates for the effect of first-time accreditation on transfers to other institutions, disaggregated by regulatory framework: (a) mandatory framework and (b) voluntary framework. The year prior to treatment ($t-1$) is the omitted reference category. Vertical bars represent 95% confidence intervals. Estimates follow Borusyak et al. (2024) with standard errors clustered at the program level.

Figure A.19: Event study estimates: permanent exits by regulatory framework.



(a) Mandatory framework (b) Voluntary framework

Note: This figure plots event study estimates for the effect of first-time accreditation on permanent exits, disaggregated by regulatory framework: (a) mandatory framework and (b) voluntary framework. The year prior to treatment ($t-1$) is the omitted reference category. Vertical bars represent 95% confidence intervals. Estimates follow Borusyak et al. (2024) with standard errors clustered at the program level.